

Stata Lecture Demo

ECON 381

Data set: wage2.dta

LING LIU



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Install Stata



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Install Stata

- Step 1:
 - UT provides Stata software download,
 - <https://webapps.utk.edu/oit/softwaredistribution/>.
 - Login use your NetID to download.

 **Office of Innovative Technologies**

Software Distribution

OIT manages several contracts to provide you with the software and tools necessary for both work and school. This software is centrally funded and available to you at no additional cost. When you log in, you will see the software available to you based on the follow criteria:

- Your affiliation (faculty, staff, or student)
- Campus (UTK, UTHSC, UTM, UTS, etc)
- Equipment purchased with university funds or another source

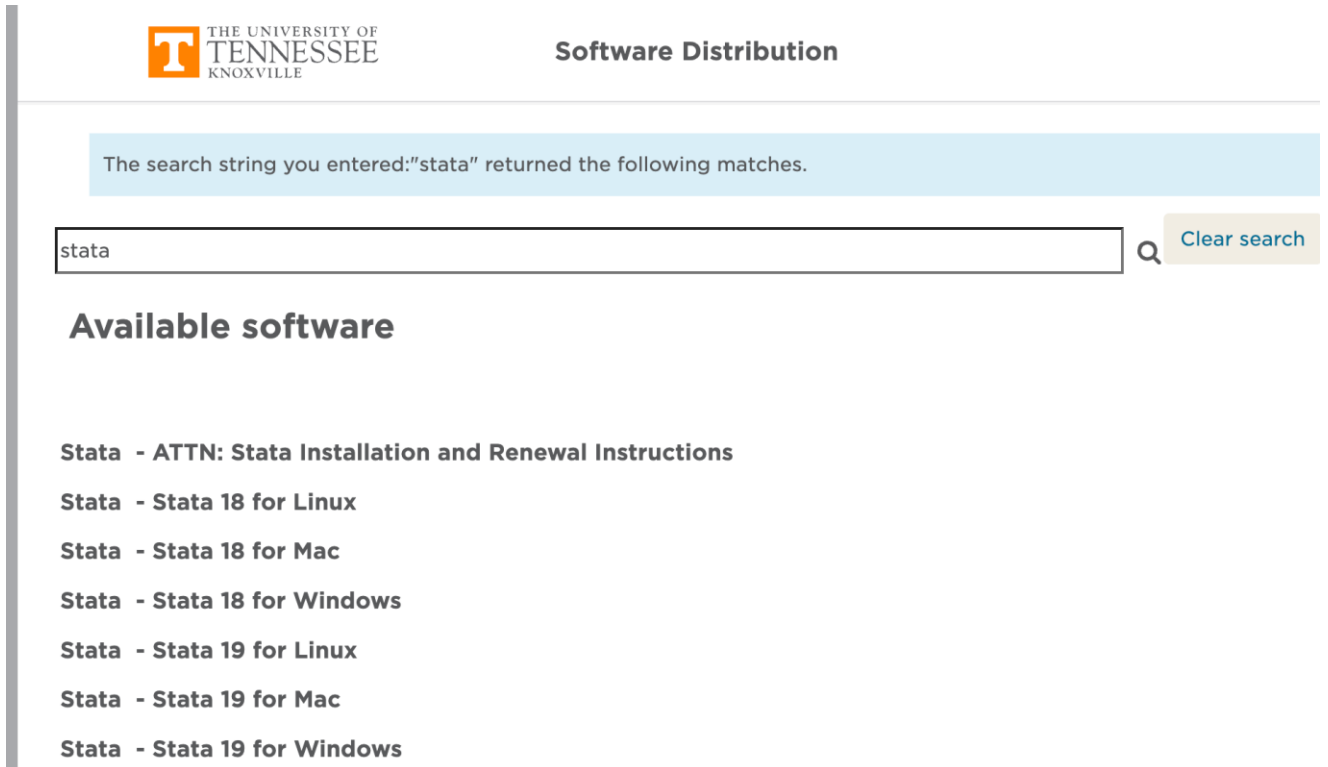
Please Note: Installation and activation instructions and codes are usually a separate download from the installer.

If you have any questions or need assistance using the site or downloading software, please [contact the OIT HelpDesk online](#) or by phone at 865-974-9900.

Log in

Install Stata

- Step 2:
 - After login, search 'Stata'
 - Pick 'Stata 19'
- **Stata 19 for Windows:** Choose this if you have a PC (Dell, HP, Lenovo, Surface, etc.).
- **Stata 19 for Mac:** Choose this if you have a MacBook, iMac, or Mac Mini. Stata 19 is fully optimized for "Apple Silicon" (M1, M2, M3, M4 chips), so it will run very fast.
- **Linux:** Only choose this if you are using an OS like Ubuntu or Fedora (usually for high-powered server research).



The screenshot shows the 'Software Distribution' page of the University of Tennessee Knoxville. At the top, the university's logo and name are displayed. Below the header, a search bar contains the text 'stata', and a 'Clear search' button is visible. A message above the search bar states: 'The search string you entered: "stata" returned the following matches.' Below the search bar, the section 'Available software' lists the following results:

- Stata - ATTN: Stata Installation and Renewal Instructions
- Stata - Stata 18 for Linux
- Stata - Stata 18 for Mac
- Stata - Stata 18 for Windows
- Stata - Stata 19 for Linux
- Stata - Stata 19 for Mac
- Stata - Stata 19 for Windows

Install Stata

- Step 2:

- After login, search 'Stata'
- Pick 'Stata 19'

- **Stata 19 for Windows:** Choose this if you have a PC (Dell, HP, Lenovo, Surface, etc.).

- **Stata 19 for Mac:** Choose this if you have a MacBook, iMac, or Mac Mini. Stata 19 is fully optimized for "Apple Silicon" (M1, M2, M3, M4 chips), so it will run very fast.

- **Linux:** Only choose this if you are using an OS like Ubuntu or Fedora (usually for high-powered server research).

Required Document

The software you have selected to download: **Stata 19 for Mac** ,
requires some additional documentation in order for you to utilize it successfully.

Step 1

Review the help document below. Save or print it. Locate any access codes, then check the box to proceed.

☒ I have reviewed the help document

Step 2

Download the software.

Please read this required document (Adobe reader required)

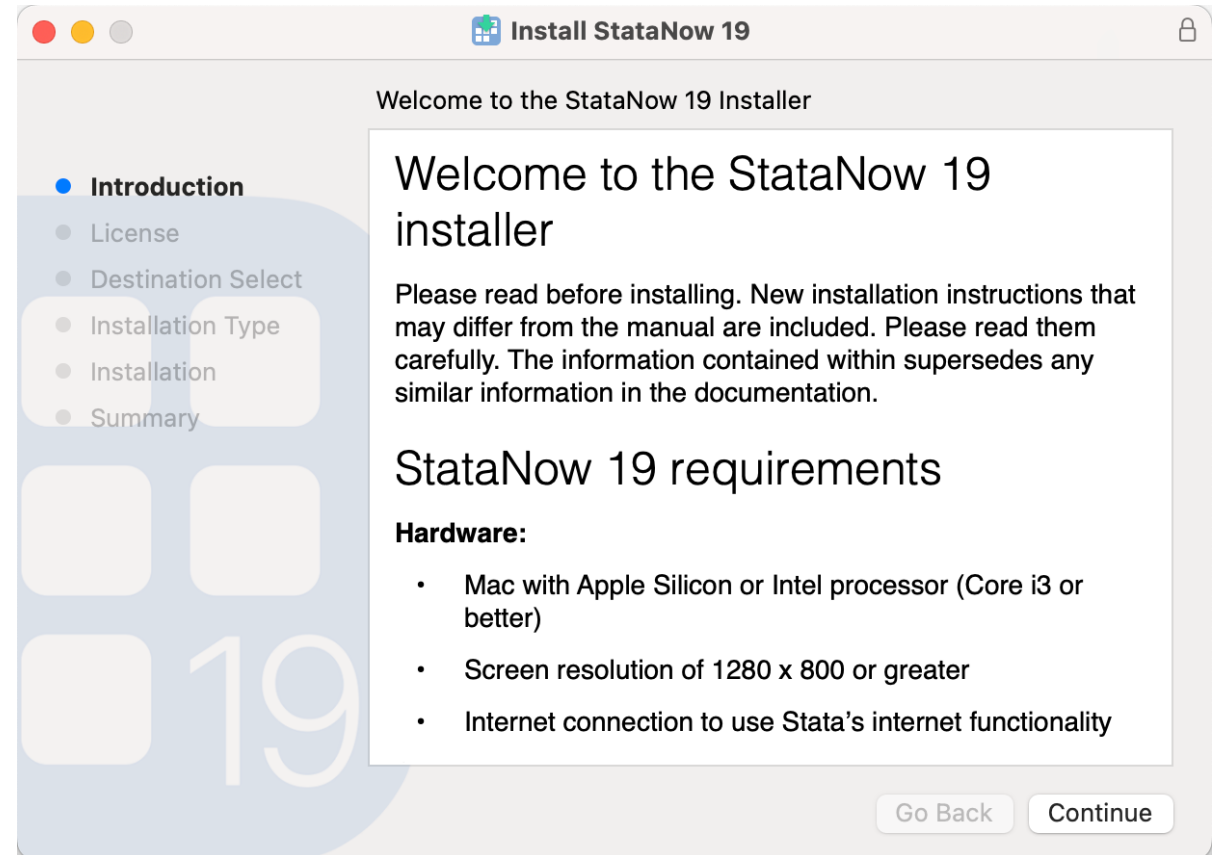
Afterwards, you may close this page and click the link in Step 2 of the main Software Distribution screen to download your install file



Install Stata

- Step 3:
 1. Double click the Stata executable file that you downloaded to begin the installation process
 - a) **Windows:** run the .exe installer
 - i. **Windows defender:** If Windows Defender SmartScreen prompts you not to run the app, click “More Info” and “Run Anyway”.
 - b) **Mac:** run the .dmg file and drag the Stata folder to Applications

Example Screenshot for Mac



Install Stata

- Step 4:
 1. Accept the license agreement and the default directory for installation.
 2. Enter your name and University of Tennessee in the Customer Information Fields.
 3. When prompted for the license type, choose “Stata/SE”.
 4. Use the default options for all other installation menu items
 5. Double click Stata to start the program
 6. When you are asked to ‘Initialize License’ for **Stata/SE**, use the this info:

WARNING: If you try to install a of Stata license other than Stata/SE, you will receive an error later, and Stata will not work on your computer. Then you will have to uninstall and reinstall Stata with the correct settings to resolve the issue.

Serial Number:	401909303029
Code:	ugu4 fnv0 ywth 5vi1 m1iq pfxf mu14 63zu nazb \$
Authorization:	kgc7

Install Stata

- Step 5:
 - Finally, enter any other additional information in required fields. When the installation is complete, the license expiration date is May 15, 2026. At that time, you can extend your license expiration for an additional year if the university decides to maintain the product.

Install Stata

If you are still not sure how to install, please check the help document in step 2 or below:

Detailed Installation Instructions

Stata has developed a detailed set of installation instructions for each operating system below:

- Windows: <https://www.stata.com/install-guide/windows/installation/>
- Mac: <https://www.stata.com/install-guide/mac/installation/>
- Linux: <https://www.stata.com/support/faqs/unix/install-download-on-linux/>
and <https://www.stata.com/manuals/ig.pdf#igInstallingStataforUnix>

How to Get Started With Stata

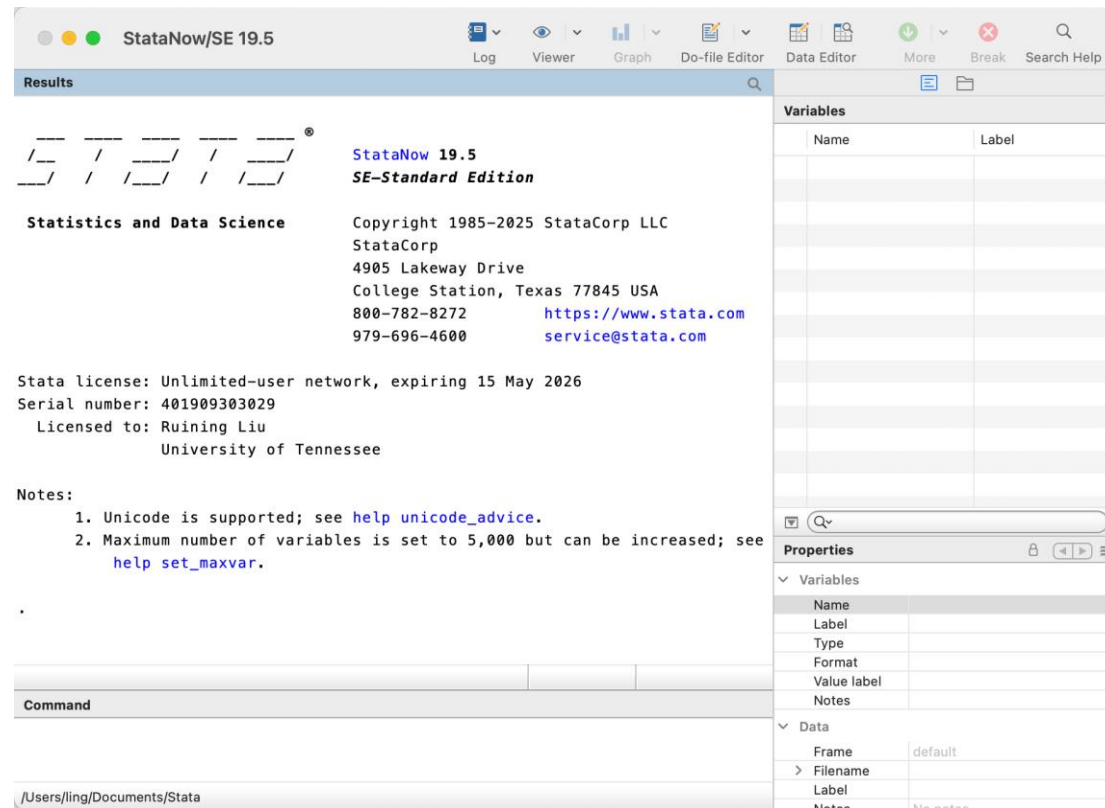


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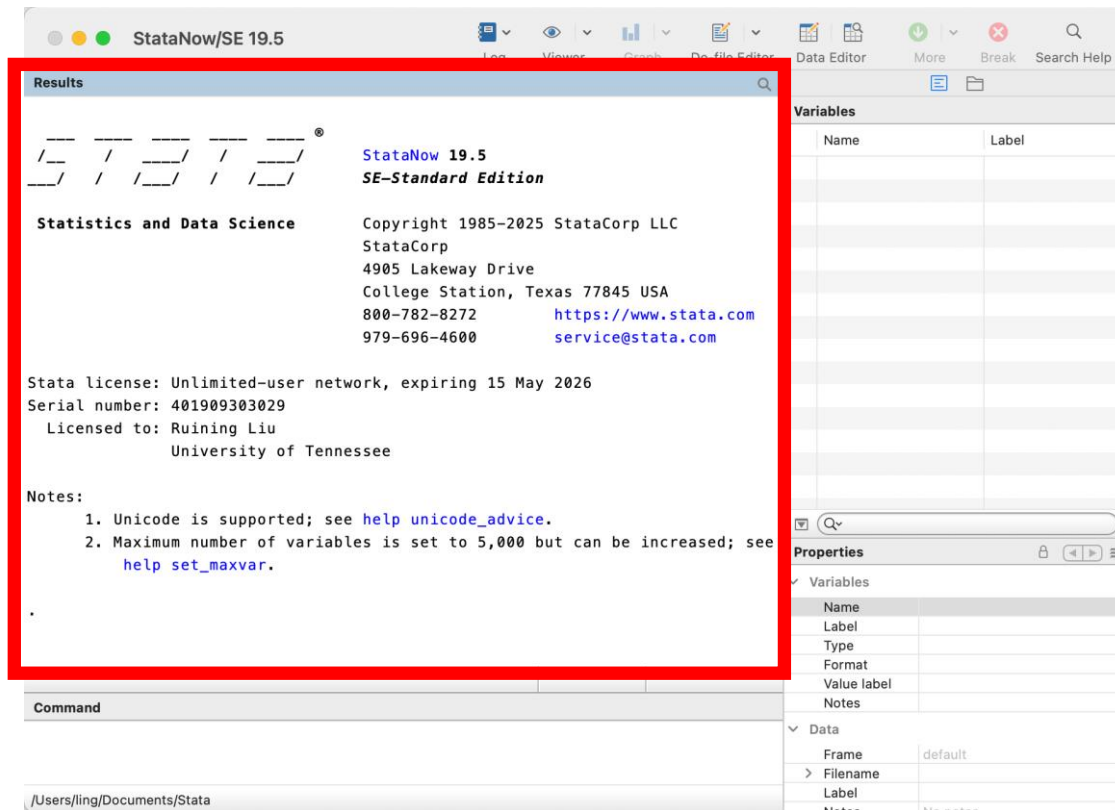
Stata Window

- This is what the Stata window looks like once opened.



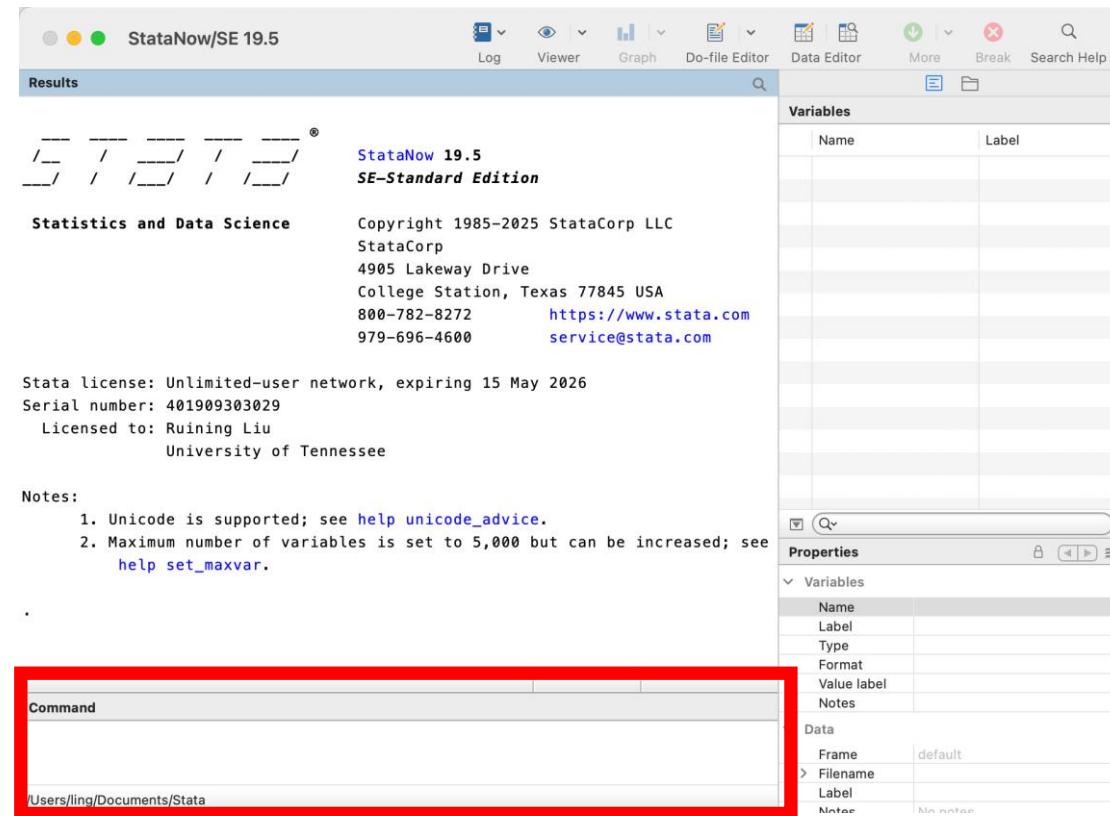
Results Window

- The results window (outlined in red) will display the results of the Stata commands you enter.



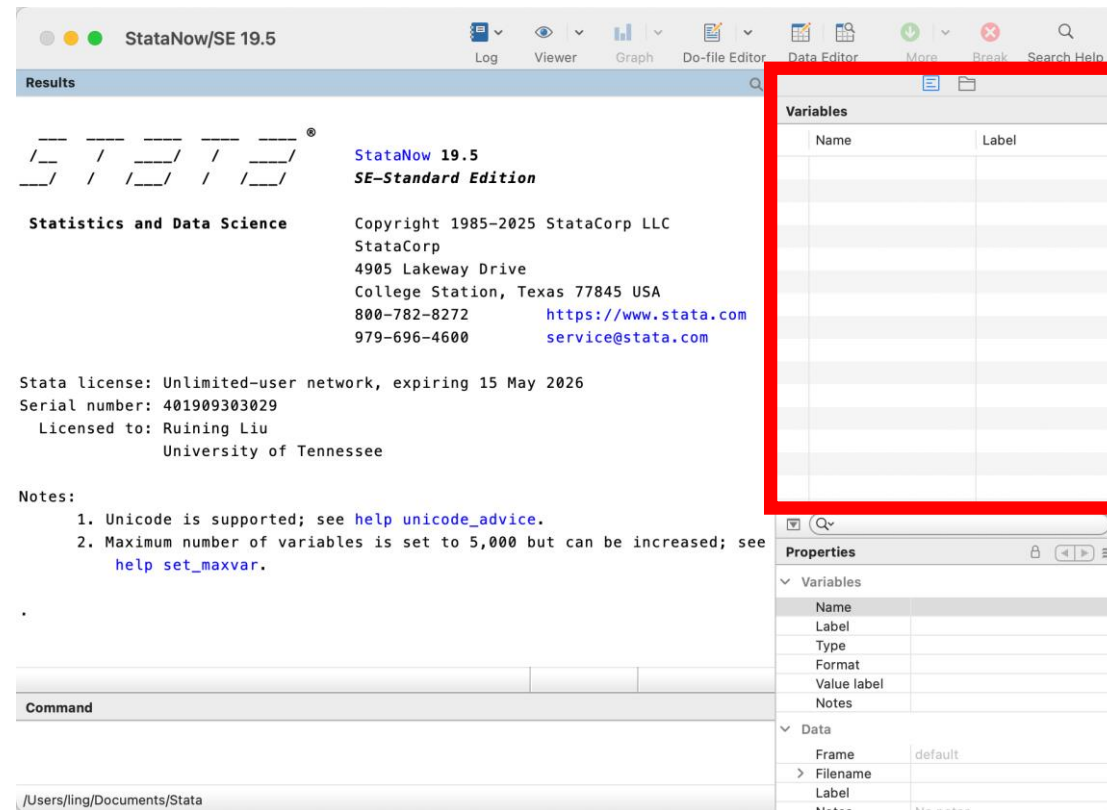
Command Window

- You can type any command into the command window, the results of which will show up in the results window.



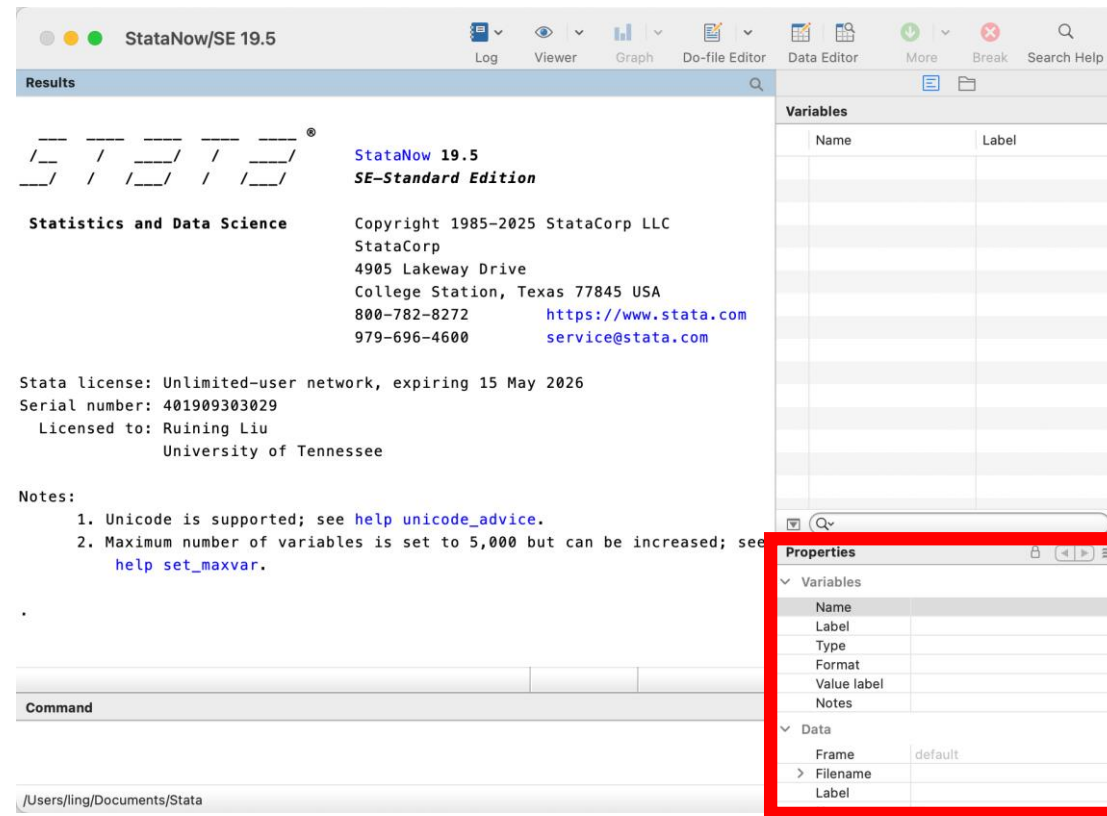
Variables Window

- Once we've loaded in a dataset, the variables and their labels will appear here.



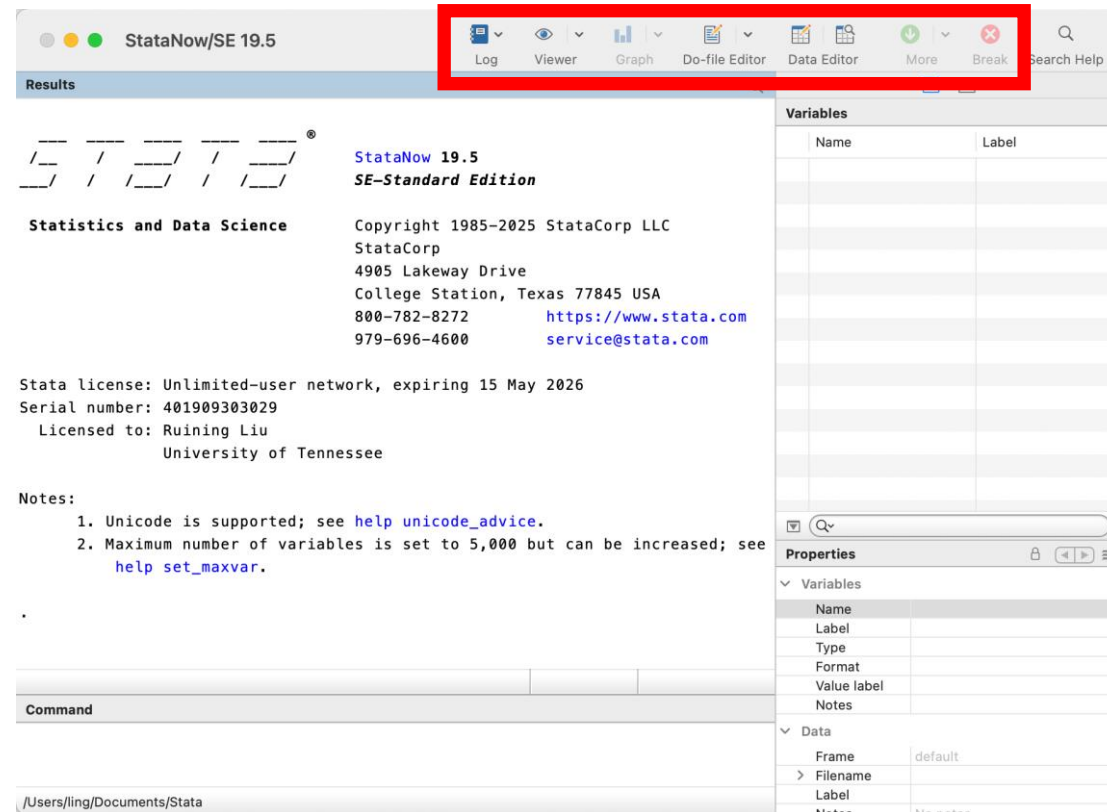
Properties window

- Additionally, the properties of our variables and the data set will appear here.



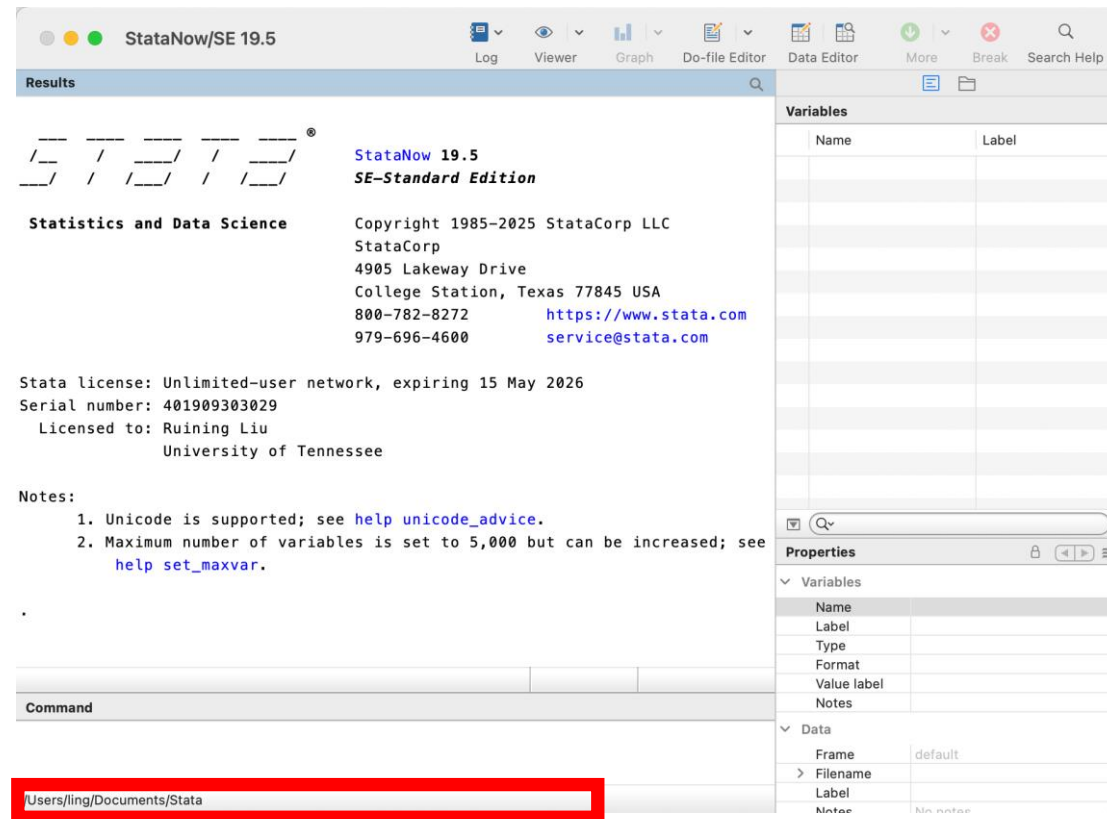
Toolbar

- The toolbar contains menu actions that are often used.



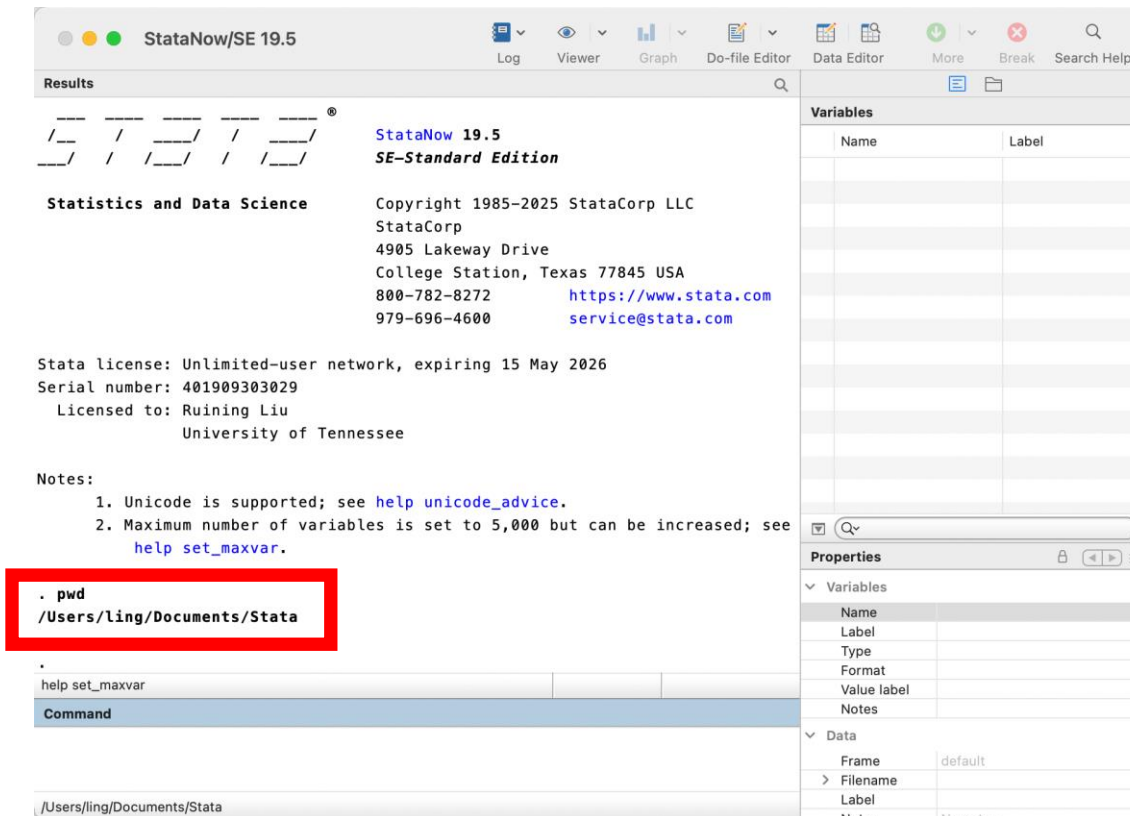
Working directory

- Underneath your command window, note that Stata prints your “working directory,” or the folder where Stata is reading files from and saving files to.



Working directory (cont'd)

- You can also view your working directory with ***pwd*** (print working directory). To do this, type ***pwd*** into the **command window** and hit enter/return.

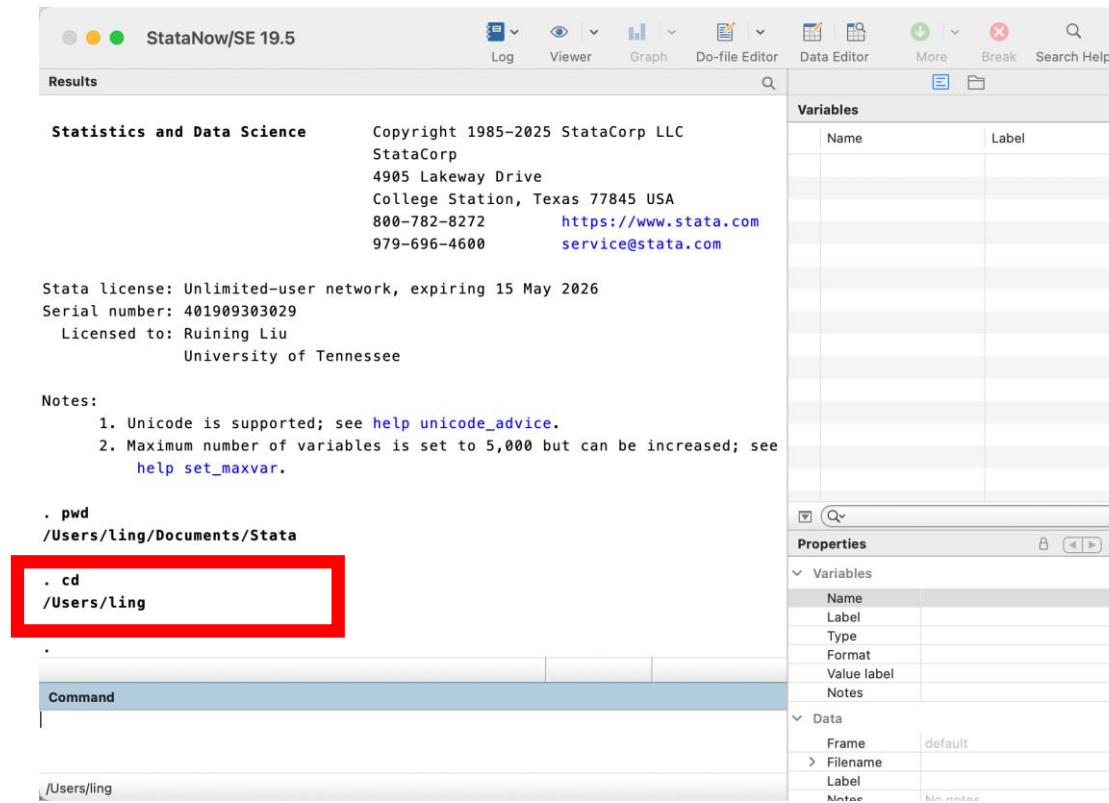


The screenshot shows the StataNow/SE 19.5 interface. The command window at the bottom displays the command `. pwd` and its output: `/Users/Ling/Documents/Stata`. The output is highlighted with a red box. The main window shows the Stata logo and version information, including the license type (Unlimited-user network) and the user (Ruining Liu) from the University of Tennessee. The right sidebar contains the Variables and Properties panels.

```
. pwd
/Users/Ling/Documents/Stata
```

Changing Directories

- To go to your home directory, use the **cd** (change directory) command. Note that your path will look different based on your operating system and file structures.



The screenshot shows the StataNow/SE 19.5 interface. The command window on the left displays the following text:

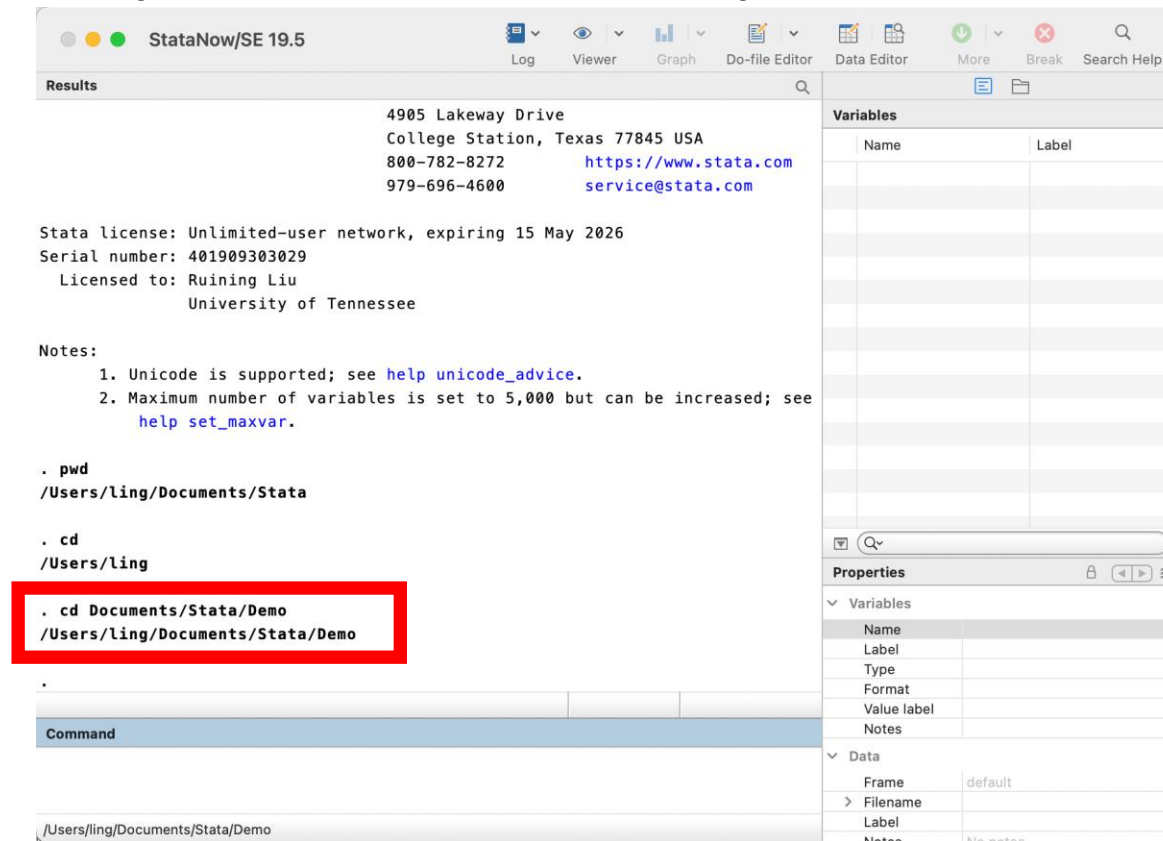
```
Statistics and Data Science      Copyright 1985-2025 StataCorp LLC  
StataCorp  
4905 Lakeway Drive  
College Station, Texas 77845 USA  
800-782-8272      https://www.stata.com  
979-696-4600      service@stata.com  
  
Stata license: Unlimited-user network, expiring 15 May 2026  
Serial number: 401909303029  
Licensed to: Ruining Liu  
University of Tennessee  
  
Notes:  
1. Unicode is supported; see help unicode_advice.  
2. Maximum number of variables is set to 5,000 but can be increased; see  
   help set_maxvar.  
  
. pwd  
/Users/ling/Documents/Stata  
  
. cd  
/Users/ling
```

The command `. cd /Users/ling` is highlighted with a red rectangle. The command window also shows the current directory as `/Users/ling` at the bottom.

The right-hand side of the interface shows the Variables and Properties panels. The Variables panel is empty, and the Properties panel shows the current variable's properties.

Changing Directories (cont'd)

- From the home directory, you can change your directory to any path you'd like. Here I went into my Documents folder, then my Stata folder, then into my Demo folder in one line.



The screenshot shows the StataNow/SE 19.5 interface. The command window on the left displays the following text:

```
4905 Lakeway Drive  
College Station, Texas 77845 USA  
800-782-8272      https://www.stata.com  
979-696-4600      service@stata.com  
  
Stata license: Unlimited-user network, expiring 15 May 2026  
Serial number: 401909303029  
Licensed to: Ruining Liu  
University of Tennessee  
  
Notes:  
1. Unicode is supported; see help unicode\_advice.  
2. Maximum number of variables is set to 5,000 but can be increased; see  
   help set\_maxvar.  
  
. pwd  
/Users/ling/Documents/Stata  
  
. cd  
/Users/ling  
  
. cd Documents/Stata/Demo  
/Users/ling/Documents/Stata/Demo  
  
Command
```

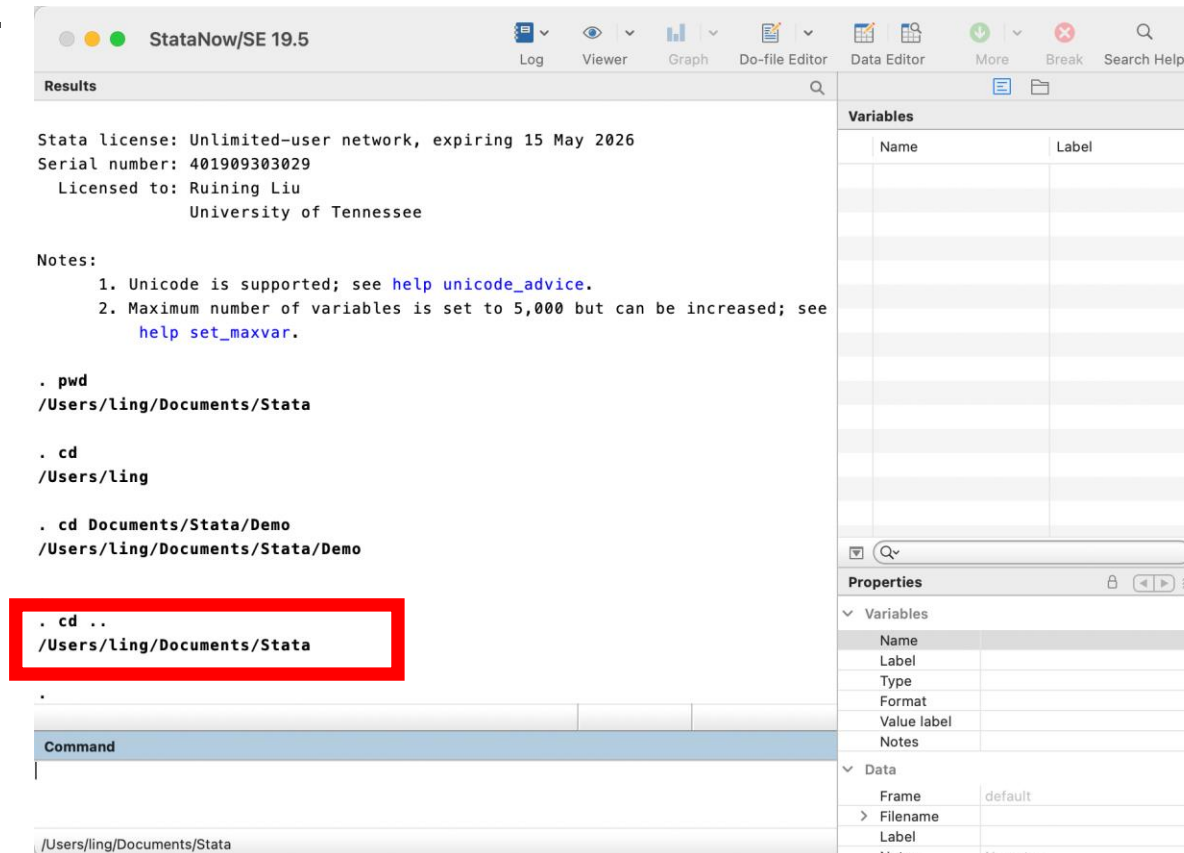
The Variables panel on the right shows a table with columns Name and Label. The Properties panel below it shows the following information:

Variables	
Name	Label

Properties	
Variables	
Name	
Label	
Type	
Format	
Value label	
Notes	
Data	
Frame	default
Filename	
Label	
Notes	No notes

Changing Directories (cont'd)

- To go back to the previous file in your path, use '**cd ..**'. If I had typed '**cd ../../**', I would be taken to /Users/ling/Documents.



The screenshot shows the StataNow/SE 19.5 interface. The main window displays the following text:

```
Stata license: Unlimited-user network, expiring 15 May 2026
Serial number: 401909303029
Licensed to: Ruining Liu
University of Tennessee

Notes:
1. Unicode is supported; see help unicode\_advice.
2. Maximum number of variables is set to 5,000 but can be increased; see
help set\_maxvar.

. pwd
/Users/ling/Documents/Stata

. cd
/Users/ling

. cd Documents/Stata/Demo
/Users/ling/Documents/Stata/Demo

. cd ..
/Users/ling/Documents/Stata
```

The last command, `. cd ..`, and its output, `/Users/ling/Documents/Stata`, are highlighted with a red rectangle. Below the command window, the current directory is shown as `/Users/ling/Documents/Stata`.

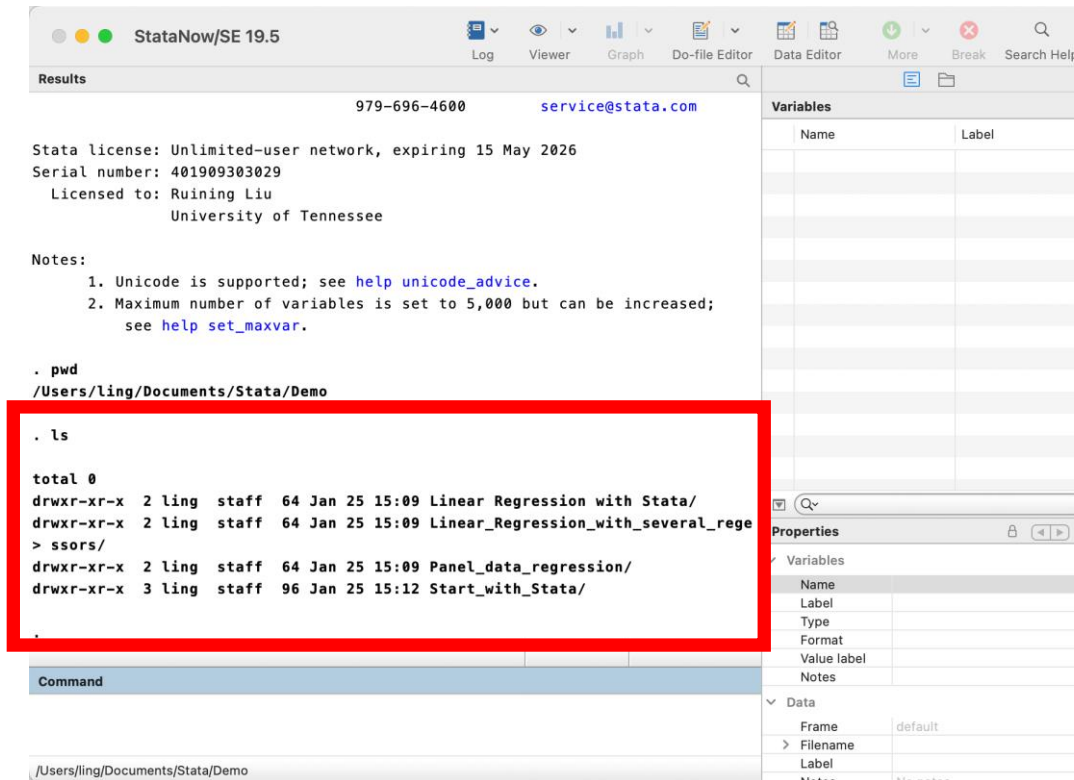
The right-hand side of the interface shows the 'Variables' and 'Properties' panels. The 'Variables' panel is empty. The 'Properties' panel shows the following details:

Variables	
Name	Label

Data	
Frame	default
Filename	
Label	
Notes	No notes

List files

- Once in a directory, to see what files and folders there are, use the **ls** (list files) command. Under my Demo folder, I have 4 subfolders as seen in the results window. Note that the **dir** command performs the same thing.



The screenshot shows the StataNow/SE 19.5 interface. The main window displays the results of the `. ls` command, which is highlighted with a red box. The results show the current directory and its contents:

```
. pwd
/Users/ling/Documents/Stata/Demo

. ls

total 0
drwxr-xr-x  2 ling  staff   64 Jan 25 15:09 Linear Regression with Stata/
drwxr-xr-x  2 ling  staff   64 Jan 25 15:09 Linear_Regression_with_several_rege
> ssors/
drwxr-xr-x  2 ling  staff   64 Jan 25 15:09 Panel_data_regression/
drwxr-xr-x  3 ling  staff   96 Jan 25 15:12 Start_with_Stata/
```

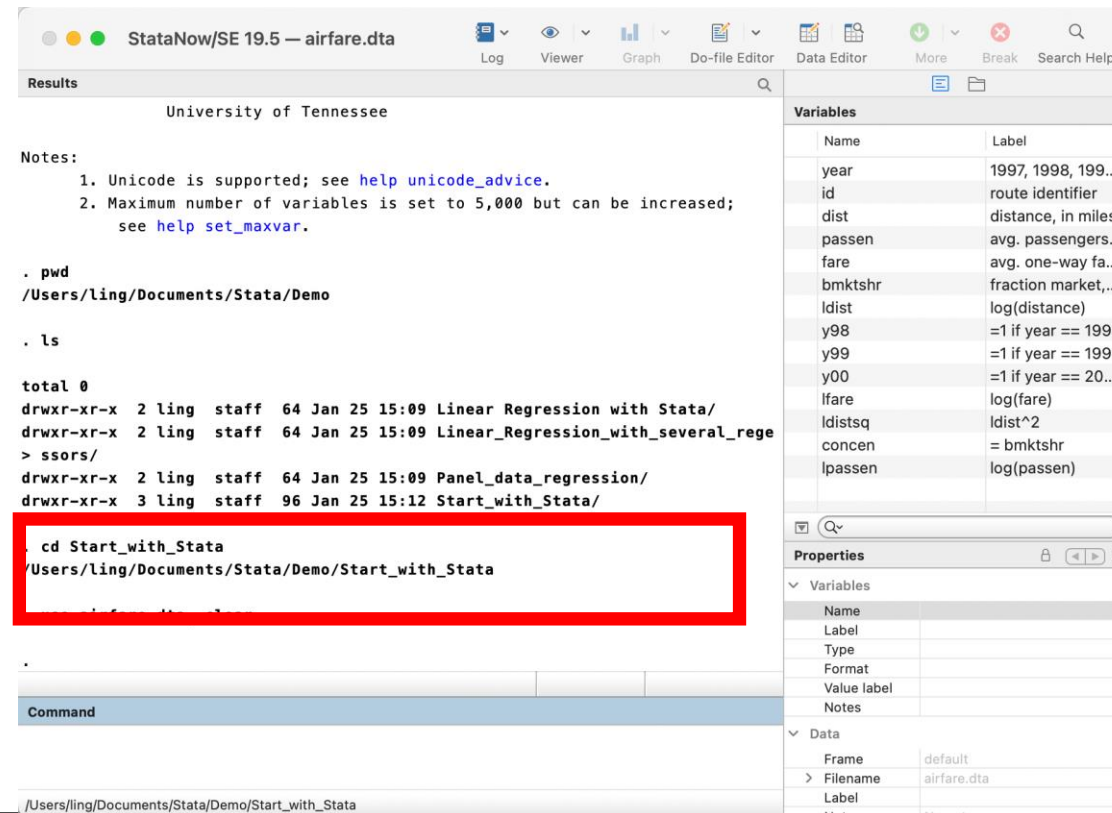
The interface also shows the Stata license information, notes, and a command window at the bottom.

Loading in data

- The .dta extension indicates a file type that is specific to Stata data sets
- Stata can only have 1 data set loaded in at a time To load in a data set, do one of two things:
 1. Double-click on the data set directly from a folder. For example, after downloading 'airfare.dta' from Canvas it should be in your Downloads folder. Double-click the .dta file from there.
 2. Navigate to the directory your data set is in using the **cd** command and then use the **use** command as shown on the next two slides.

Loading in data (cont'd)

- In this line, I navigate to the directory where I have my .dta file. Note that if your folder names contain spaces, you will have to type your path in double quotes.



The screenshot shows the StataNow/SE 19.5 interface with the file 'airfare.dta' open. The command window on the left displays the following commands and output:

```
. pwd
/Users/ling/Documents/Stata/Demo

. ls

total 0
drwxr-xr-x  2 ling  staff  64 Jan 25 15:09 Linear_Regression_with_Stata/
drwxr-xr-x  2 ling  staff  64 Jan 25 15:09 Linear_Regression_with_several_rege
> ssors/
drwxr-xr-x  2 ling  staff  64 Jan 25 15:09 Panel_data_regression/
drwxr-xr-x  3 ling  staff  96 Jan 25 15:12 Start_with_Stata/

. cd Start_with_Stata
/Users/ling/Documents/Stata/Demo/Start_with_Stata
```

The command `. cd Start_with_Stata` and its output are highlighted with a red box. The right-hand side of the interface shows the 'Variables' list and the 'Properties' panel.

Name	Label
year	1997, 1998, 199...
id	route identifier
dist	distance, in miles
passen	avg. passengers...
fare	avg. one-way fa...
bmktshr	fraction market,...
ldist	log(distance)
y98	=1 if year == 1998
y99	=1 if year == 1999
y00	=1 if year == 20...
lfare	log(fare)
ldistsq	ldist^2
concen	= bmktshr
lpassen	log(passen)

Properties	
Variables	
Name	
Label	
Type	
Format	
Value label	
Notes	
Data	
Frame	default
Filename	airfare.dta
Label	
Notes	

Loading in data (cont'd)

- To load in the data, the syntax is '**use filename, clear**'. The clear option clears out any existing data file loaded into Stata. Now that we have loaded in data, the variables window has populated with the variable names and corresponding labels.

The screenshot displays the Stata 19.5 interface. The command window on the left shows the following commands and output:

```
. pwd
/Users/Ling/Documents/Stata/Demo/Start_with_Stata

. use WAGE2.DTA, clear
```

The output of the `pwd` command is `/Users/Ling/Documents/Stata/Demo/Start_with_Stata`. The `use WAGE2.DTA, clear` command is highlighted with a red box.

The Variables window on the right, also highlighted with a red box, lists the following variables and their labels:

Name	Label
wage	average hourly earnings
educ	years of education
exper	years potential experience
tenure	years with current employer
nonwhite	=1 if nonwhite
female	=1 if female
married	=1 if married
numdep	number of dependents
smsa	=1 if live in SMSA
northcen	=1 if live in north central U.S.
south	=1 if live in southern region
west	=1 if live in western region
construc	=1 if work in construc. indus.
ndurman	=1 if in nondur. manuf. indus.
trcommu	=1 if in trans, commun, pub ut

The bottom of the Variables window shows the Data section with the following information:

Frame	default
Filename	WAGE2.DTA
Label	
Notes	

Data description (wage2.dta)

- **Source:** Jeffrey M. Wooldridge, *Introductory Econometrics: A Modern Approach*.
- **Original Study:** Cross-sectional data from the 1976 Current Population Survey (CPS).
- **Structure:**
 - **Dataset Identification:** Although our working file is named wage2.dta, it corresponds to the **wage1** dataset in the Wooldridge textbook (based on the 1976 Current Population Survey).
 - **Observations:** **526** individuals.
 - **Variables:** 24 variables covering earnings, education, and demographics.
- **Purpose:** This dataset is the standard "workhorse" for learning labor economics and Mincerian wage regressions (estimating the impact of education and experience on earnings).

Data description (cont'd)

- This dataset is a classic labor economics cross-section of individual workers. Each row represents a unique individual, and the columns capture their earnings, education, and demographic characteristics.

Variable Name	Label / Description	Economic Intuition
wage	Average hourly earnings	The dependent variable (outcome) we want to explain.
educ	Years of education	The primary independent variable (human capital).
exper	Years of potential work experience	Controls for "learning by doing" in the labor market.
tenure	Years with current employer	Measures firm-specific human capital or seniority.

Data description (cont'd)

- In this dataset, many variables are **binary indicators**. They act like a "light switch" to categorize individuals into two distinct groups.
- A value of **1** means the condition is **TRUE** (Yes), and a value of **0** means the condition is **FALSE** (No).

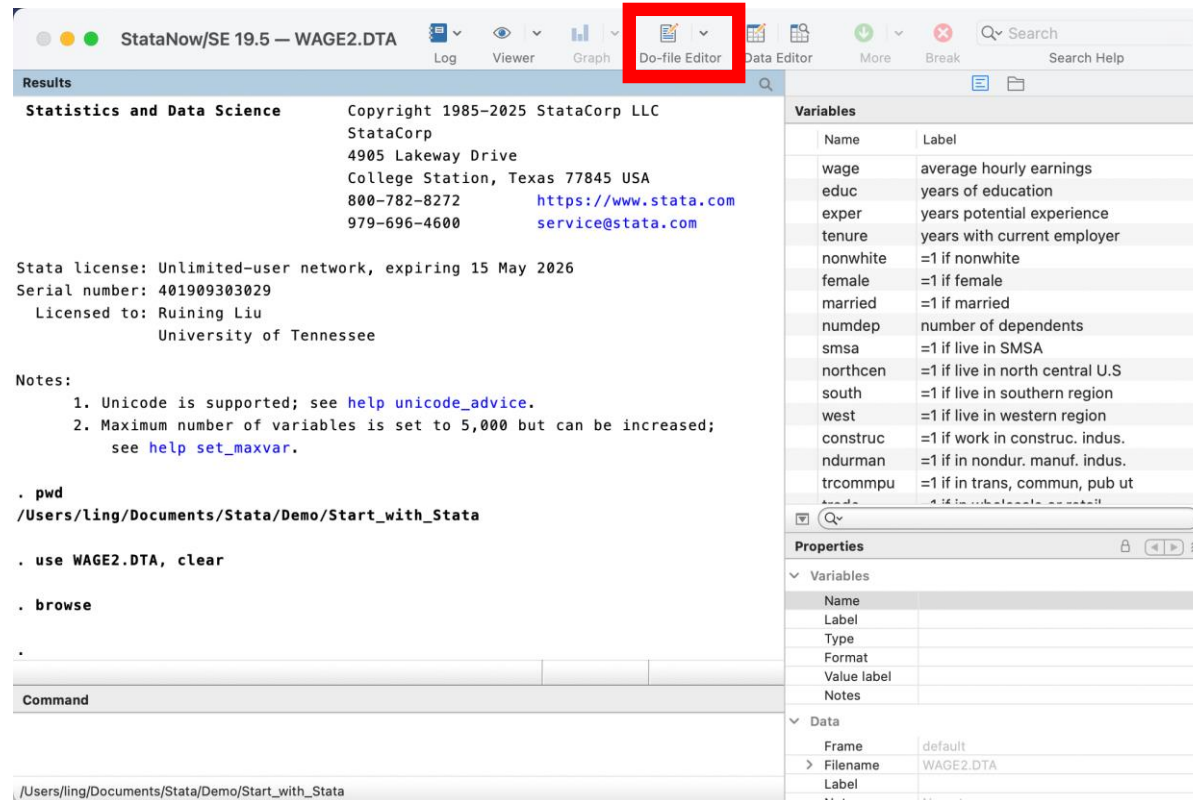
Variable	Description (Value = 1)
female	The individual is female.
married	The individual is currently married.
nonwhite	The individual is non-white.
smsa	The individual lives in an urban area (Standard Metropolitan Statistical Area).
south / west	The individual lives in that specific geographic region.

Do-files

- When we code something in Stata that we will want to come back to or send to others (i.e., turning in your homework), we will need a separate file that contains the code. Coding directly into the command window will not save the code anywhere for us.
- This is what we use do-files for. Think of the do-file as a text editor that we can run Stata code out of. The do-file is the equivalent of a script in other programming languages.
- As with .dta, the .do extension is specific to Stata.

Starting a do-file

- To open a do-file, click the icon in the toolbar that says 'Do-file Editor'. A new window will appear.



Starting a do-file (cont'd)

- Now we have both the main window (left) and the do-file (right) open.

The screenshot displays the StataNow/SE 19.5 interface. The main window on the left shows the 'Results' tab with the following content:

Statistics and Data Science Copyright 1985–2025 StataCorp LLC
StataCorp
4905 Lakeway Drive
College Station, Texas 77845 USA
800-782-8272 <https://www.stata.com>
979-696-4600 service@stata.com

Stata license: Unlimited-user network, expiring 15 May 2026
Serial number: 401909303029
Licensed to: Ruining Liu
University of Tennessee

Notes:

1. Unicode is supported; see [help unicode_advice](#).
2. Maximum number of variables is set to 5,000 but can be increased; see [help set_maxvar](#).

```
. pwd  
/Users/ling/Documents/Stata/Demo/Start_with_Stata  
  
. use WAGE2.DTA, clear  
  
. browse  
  
.
```

The 'Command' window at the bottom is empty.

The 'Variables' window on the right shows a list of variables with their labels:

Name	Label
wage	average hourly earning
educ	years of education
exper	years potential experie
tenure	years with current emp
nonwhite	=1 if nonwhite
female	=1 if female
married	=1 if married
numdep	number of dependents
smsa	=1 if live in SMSA
northcen	=1 if live in north centr
south	=1 if live in southern re
west	=1 if live in western reg
construc	=1 if work in construc.
ndurman	=1 if in nondur. manuf.
trcompu	=1 if in trans, commun

The 'Properties' window shows the following details for the selected variable:

Name	Label	Type	Format	Value label	Notes

The 'Data' window shows the first row of data:

1

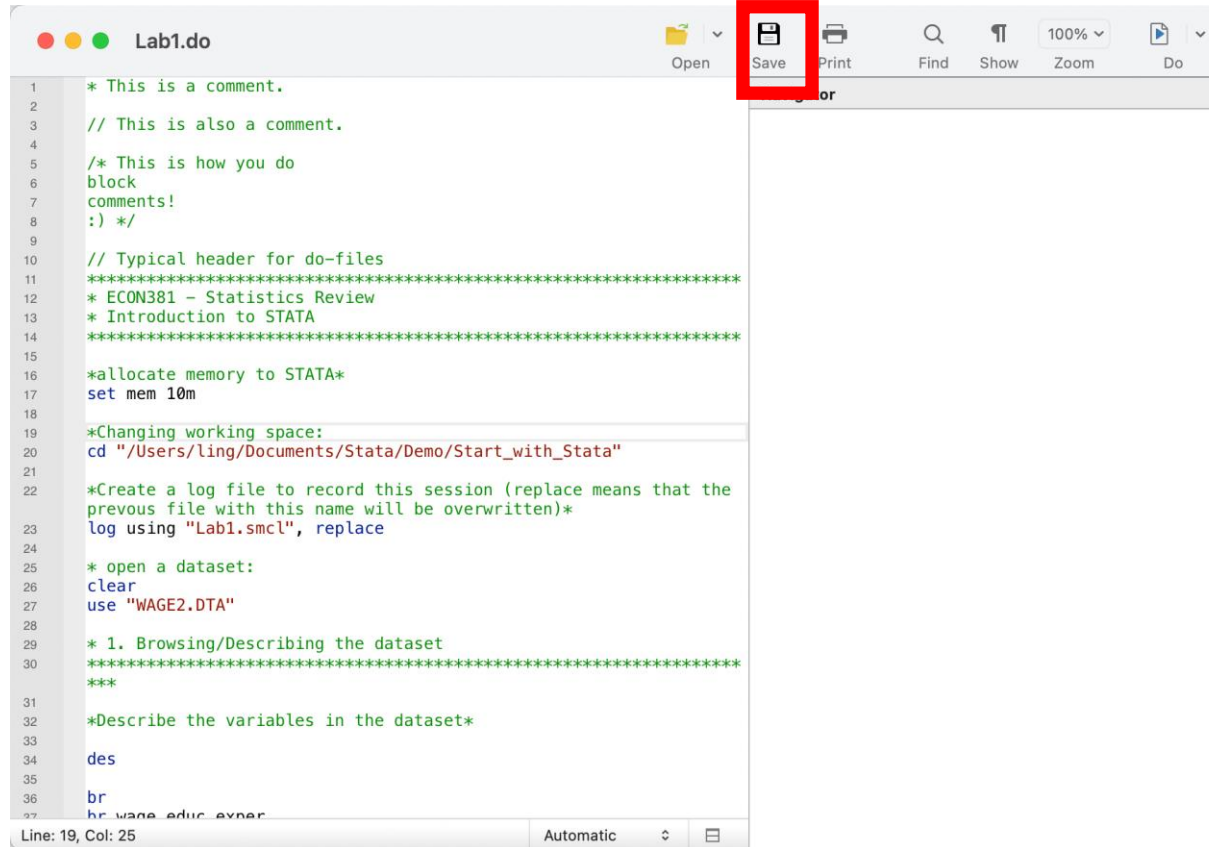
The 'Untitled' do-file editor window on the right is open and empty, with a status bar indicating 'Line: 1, Col: 1'.

Commenting

- It is best practice in any programming language to write comments along with your code. This will improve the readability for both those you may share your code with, and you in the future.
- You can type single-line comments using either '*' or `//` in your do-file. You can also type multi-line comments, where anything you type in between `/*` and `*/` will appear as a comment. Examples of these comments are shown on the next slide.

Commenting and saving

- A common header for a do-file is shown, including the title, author, and date. Additionally, you should save your do-file using the save icon in the toolbar of the do-file. I saved mine and named it 'Lab1.do'.



```
1  * This is a comment.
2
3  // This is also a comment.
4
5  /* This is how you do
6  block
7  comments!
8  :) */
9
10 // Typical header for do-files
11 *****
12 * ECON381 - Statistics Review
13 * Introduction to STATA
14 *****
15
16 *allocate memory to STATA*
17 set mem 10m
18
19 *Changing working space:
20 cd "/Users/ling/Documents/Stata/Demo/Start_with_Stata"
21
22 *Create a log file to record this session (replace means that the
23 previous file with this name will be overwritten)*
24 log using "Lab1.smcl", replace
25
26 * open a dataset:
27 clear
28 use "WAGE2.DTA"
29
30 * 1. Browsing/Describing the dataset
31 *****
32 ***
33
34 *Describe the variables in the dataset*
35
36 des
37
38 br
39 hr wage educ exper
```

Executing commands

- To execute a command from the do-file, begin by typing the command and then highlighting it. To have it run, click the 'Do' button in the top right. The results will show in the results window. By clicking 'Do' without highlighting a specific line, Stata will run the entire do-file. Here, I am attempting to calculate 3+3.

The screenshot displays the Stata 19.5 interface with three main windows: Results, Variables, and a do-file editor (Lab1.do).

Results Window: Shows the command log. The command `. 3+3` is highlighted in red, with an error message below it: `3 is not a valid command name`. The command log also shows the path `/Users/ling/Documents/Stata/Demo/Start_with_Stata` and the command `. use WAGE2.DTA, clear`.

Variables Window: Lists variables from the WAGE2.DTA dataset. The variables are: wage (average hourly earning), educ (years of education), exper (years potential experience), tenure (years with current employer), nonwhite (1 if nonwhite), female (1 if female), married (1 if married), numdep (number of dependents), smsa (1 if live in SMSA), northcen (1 if live in north central), south (1 if live in southern region), west (1 if live in western region), construc (1 if work in construction), ndurman (1 if in nondurable manufacturing), and trcommpu (1 if in transportation, communication, and utilities).

Lab1.do Editor: Shows the source code. The line `display 3+3` is highlighted in blue. The 'Do' button in the top right corner of the editor is highlighted in red.

Executing commands

- Unfortunately, Stata's command window cannot be used as a calculator as in other programming languages. To have Stata compute a mathematical expression, we use the display command as shown below.

The screenshot shows the Stata 19.5 interface with a do-file named 'Lab1.do' open. The 'Results' window on the left shows the execution of the do-file, including the command `. display 3+3` which resulted in the output `6`. A red box highlights the command `. display 3+3 // use the display command to compute mathematical expressions` in the do-file editor. The 'Variables' window in the center lists variables such as 'wage', 'educ', 'exper', 'tenure', 'nonwhite', 'female', 'married', 'numdep', 'smsa', 'northcen', 'south', 'west', 'construc', 'ndurman', and 'trcommu'. The 'Properties' window on the right shows the properties of the selected variable, including Name, Label, Type, Format, Value label, and Notes. The 'Data' window at the bottom shows the data frame.

```
. 3+3 // typing mathematical expressions will result in an error
3 is not a valid command name
r(199);

end of do-file

r(199);

. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. display 3+3 // use the display command to compute mathematical expressions
6

.
end of do-file

. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. display 3+3 // use the display command to compute mathematical expressions
6

.
end of do-file
```

```
* This is a comment.
// This is also a comment.
/* This is how you do
block
comments!
:) */

// Typical header for do-files
*****
* ECON381 – Statistics Review
* Introduction to STATA
*****

// Let's learn some Stata commands:

3+3 // typing mathematical expressions will result in an error
display 3+3 // use the display command to compute mathematical
expressions

*allocate memory to STATA*
set mem 10m

*Changing working space:
cd "/Users/ling/Documents/Stata/Demo/Start_with_Stata"

*Create a log file to record this session (replace means that the
previous file with this name will be overwritten)*
log using "Lab1.smcl", replace

* open a dataset:
clear
use "WAGE2.DTA"

* 1. Browsing/Describing the dataset
*****
```

Data description

- To have description of the loaded dataset, use the **describe/des** command.

The screenshot displays the Stata software interface with two windows open: 'StataNow/SE 19.5 — WAGE2.DTA' and 'Lab1.do'.

In the 'StataNow/SE 19.5 — WAGE2.DTA' window, the command window shows the following code:

```
. use "WAGE2.DTA"
.
. end of do-file
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"
. des
```

The 'Results' window shows the output of the 'des' command:

```
Contains data from WAGE2.DTA
Observations:      526
Variables:         24      16 Sep 1996 15:52
```

Variable name	Storage type	Display format	Value label	Variable label
wage	float	%8.2g		average hourly earnings
educ	byte	%8.0g		years of education
exper	byte	%8.0g		years potential experience
tenure	byte	%8.0g		years with current employer
nonwhite	byte	%8.0g		=1 if nonwhite
female	byte	%8.0g		=1 if female
married	byte	%8.0g		=1 if married
numdep	byte	%8.0g		number of dependents
smsa	byte	%8.0g		=1 if live in SMSA
northcen	byte	%8.0g		=1 if live in north central U.S.
south	byte	%8.0g		=1 if live in southern region

The 'Lab1.do' window shows the following code:

```
*Create a log file to record this session (replace means that the
previous file with this name will be overwritten)*
log using "Lab1.smcl", replace

* open a dataset:
clear
use "WAGE2.DTA"

* 1. Browsing/Describing the dataset
*****
*Describe the variables in the dataset*
des

br wage educ exper

*Gives summary statistics of the variables wage education*
sum wage educ

sum wage, detail
sum educ, detail

*list observations 1 through 15*
list wage educ in 1/15

*Tabulate Distribution of Education *
tab educ
tab exper

* Sorting the data:
sort wage

* Using "in" and "if"
browse in 1/10
browse if educ <= 12
browse if educ <= 12 & exper >= 10
browse if educ <= 12 & ( exper <= 2 | exper >= 20 )

* Create histograms:
twoway (histogram wage)
twoway (histogram educ)
```

The 'Variables' window shows a list of variables with their names and labels:

Name	Label
wage	average hourly earnings
educ	years of education
exper	years potential experience
tenure	years with current employer
nonwhite	=1 if nonwhite
female	=1 if female
married	=1 if married
numdep	number of dependents
smsa	=1 if live in SMSA
northcen	=1 if live in north central U.S.
south	=1 if live in southern region

The 'Properties' window shows the following information:

Property	Value
Variables	
Name	
Label	
Type	
Format	
Value label	
Notes	

The 'Data' window shows the following information:

Property	Value
Frame	default
Filename	WAGE2.DTA
Label	

Browsing data

- To browse the .dta file, use the **browse/br** command or click the data icon with the magnifying glass in the toolbar. A new window will appear.

The screenshot displays the StataNow/SE 19.5 interface. The main window shows the command window with the following content:

```
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"
. br
. end of do-file
```

The command `. br` is highlighted with a red box. The results window on the left shows the following data:

Variable	Type	Format	Label
south	byte	%8.0g	=1 if live in southern region
west	byte	%8.0g	=1 if live in western region
construc	byte	%8.0g	=1 if work in construc. indus.
ndurman	byte	%8.0g	=1 if in nondur. manuf. indus.
trcommpu	byte	%8.0g	=1 if in trans, commun, pub ut
trade	byte	%8.0g	=1 if in wholesale or retail
services	byte	%8.0g	=1 if in services indus.
profserv	byte	%8.0g	=1 if in prof. serv. indus.
profocc	byte	%8.0g	=1 if in profess. occupation
clerocc	byte	%8.0g	=1 if in clerical occupation
servocc	byte	%8.0g	=1 if in service occupation
lwage	float	%9.0g	log(wage)
expersq	int	%9.0g	exper^2
tenursq	int	%9.0g	tenure^2

The command window on the right, titled 'Lab1.do', contains the following code:

```
*Create a log file to record this session (replace means that the
previous file with this name will be overwritten)*
log using "Lab1.smcl", replace

* open a dataset:
clear
use "WAGE2.DTA"

* 1. Browsing/Describing the dataset
*****
***

*Describe the variables in the dataset*
des

br
br wage educ exper

*Gives summary statistics of the variables wage education*
sum wage educ

sum wage, detail
sum educ, detail

*list observations 1 through 15*
list wage educ in 1/15

*Tabulate Distribution of Education *
tab educ
tab exper

* Sorting the data:
sort wage

* Using "in" and "if"
browse in 1/10
browse if educ <= 12
browse if educ <= 12 & exper >= 10
browse if educ <= 12 & ( exper <= 2 | exper >= 20 )

* Create histograms:
twoway (histogram wage)
twoway (histogram educ)
```

The command `br` is highlighted with a red box. The status bar at the bottom indicates 'Line: 42, Col: 3' and 'Automatic'.

Browsing data (cont'd)

- This is a viewable version of the WAGE2.DTA file. Each row corresponds to an individual observation (a worker), and each column corresponds to a variable. For example, **row 1** corresponds to all the attributes of the first individual in the dataset. This particular person has an average hourly wage of **3.1** (**wage** variable) and has completed **11** years of schooling (**educ** variable).

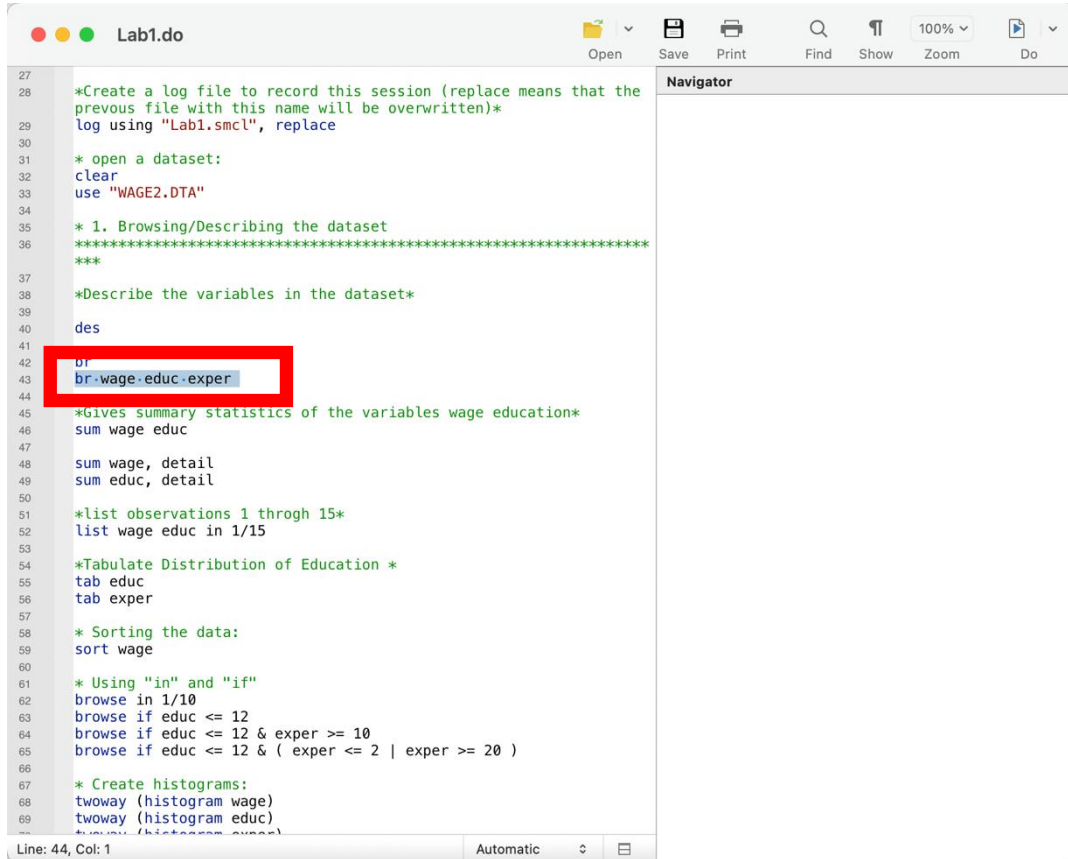
Data Editor (Browse) — WAGE2.DTA

Vars: 24 Order: Dataset Obs: 526

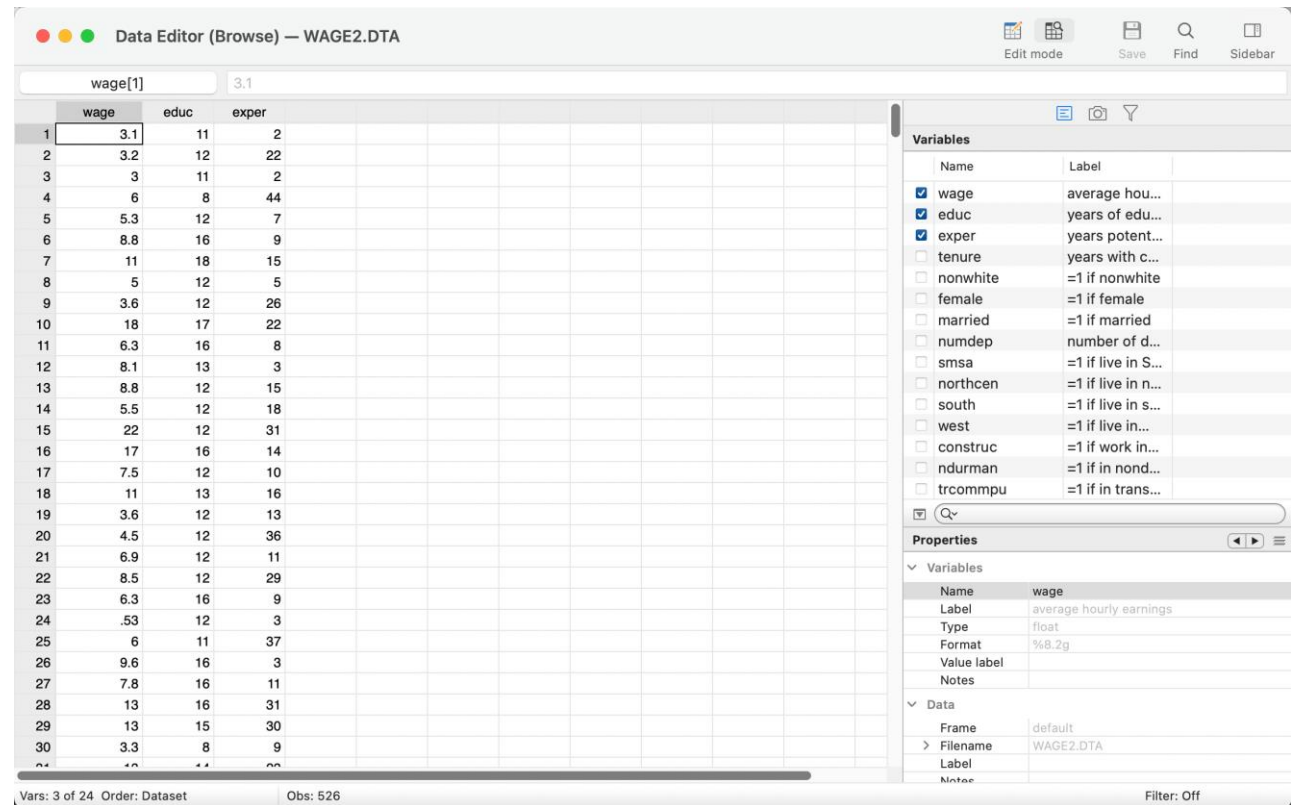
	wage	educ	exper	tenure	nonwhite	female	married	numdep	smsa	northcen	south	west	construc	ndurman	trcompu
1	3.1	11	2	0	0	1	0	2	1	0	0	0	0	0	0
2	3.2	12	22	2	0	1	1	3	1	0	0	0	0	0	0
3	3	11	2	0	0	0	0	2	0	0	0	0	0	0	0
4	6	8	44	28	0	0	1	0	1	0	0	0	0	0	0
5	5.3	12	7	2	0	0	1	1	0	0	0	0	0	0	0
6	8.8	16	9	8	0	0	1	0	1	0	0	0	0	0	0
7	11	18	15	7	0	0	0	0	1	0	0	0	0	0	0
8	5	12	5	3	0	1	0	0	1	0	0	0	0	0	0
9	3.6	12	26	4	0	1	0	2	1	0	0	0	0	0	0
10	18	17	22	21	0	0	1	0	1	0	0	0	0	0	0
11	6.3	16	8	2	0	1	0	0	1	0	0	0	0	0	0
12	8.1	13	3	0	0	1	0	0	1	0	0	0	0	0	0
13	8.8	12	15	0	0	0	1	2	1	0	0	0	0	0	0
14	5.5	12	18	3	0	0	0	0	1	0	0	0	0	0	0
15	22	12	31	15	0	0	1	1	1	0	0	0	0	0	0
16	17	16	14	0	0	0	1	1	1	0	0	0	0	0	0
17	7.5	12	10	0	0	1	1	0	1	0	0	0	0	0	0
18	11	13	16	10	0	1	0	0	1	0	0	0	0	0	0
19	3.6	12	13	0	0	1	1	3	1	0	0	0	0	0	0
20	4.5	12	36	6	0	1	1	0	1	0	0	0	0	0	0
21	6.9	12	11	4	0	1	0	0	1	0	0	0	0	0	0
22	8.5	12	29	13	0	0	1	3	1	0	0	0	0	0	0
23	6.3	16	9	9	0	1	0	0	1	0	0	0	0	0	0
24	.53	12	3	1	0	1	0	0	1	0	0	0	0	0	0
25	6	11	37	8	1	1	0	0	1	0	0	0	0	0	0
26	9.6	16	3	3	1	0	1	1	1	0	0	0	0	0	0
27	7.8	16	11	10	0	0	1	1	1	0	0	0	0	0	0
28	13	16	31	0	0	0	1	0	1	0	0	0	0	0	0
29	13	15	30	0	0	0	1	2	1	0	0	0	0	0	0
30	3.3	8	9	1	0	1	1	2	1	0	0	0	0	0	0

Browsing data (cont'd)

- You can also browse certain columns, use the ***browse/br name1 name2 name3*** command.



```
27 *Create a log file to record this session (replace means that the
28 previous file with this name will be overwritten)*
29 log using "Lab1.smcl", replace
30
31 * open a dataset:
32 clear
33 use "WAGE2.DTA"
34
35 * 1. Browsing/Describing the dataset
36 *****
37
38 *Describe the variables in the dataset*
39 des
40
41 br wage educ exper
42
43 *Gives summary statistics of the variables wage education*
44 sum wage educ
45
46 sum wage, detail
47 sum educ, detail
48
49 *list observations 1 through 15*
50 list wage educ in 1/15
51
52 *Tabulate Distribution of Education *
53 tab educ
54 tab exper
55
56 * Sorting the data:
57 sort wage
58
59 * Using "in" and "if"
60 browse in 1/10
61 browse if educ <= 12
62 browse if educ <= 12 & exper >= 10
63 browse if educ <= 12 & ( exper <= 2 | exper >= 20 )
64
65 * Create histograms:
66 twoway (histogram wage)
67 twoway (histogram educ)
68 twoway (histogram exper)
69
```



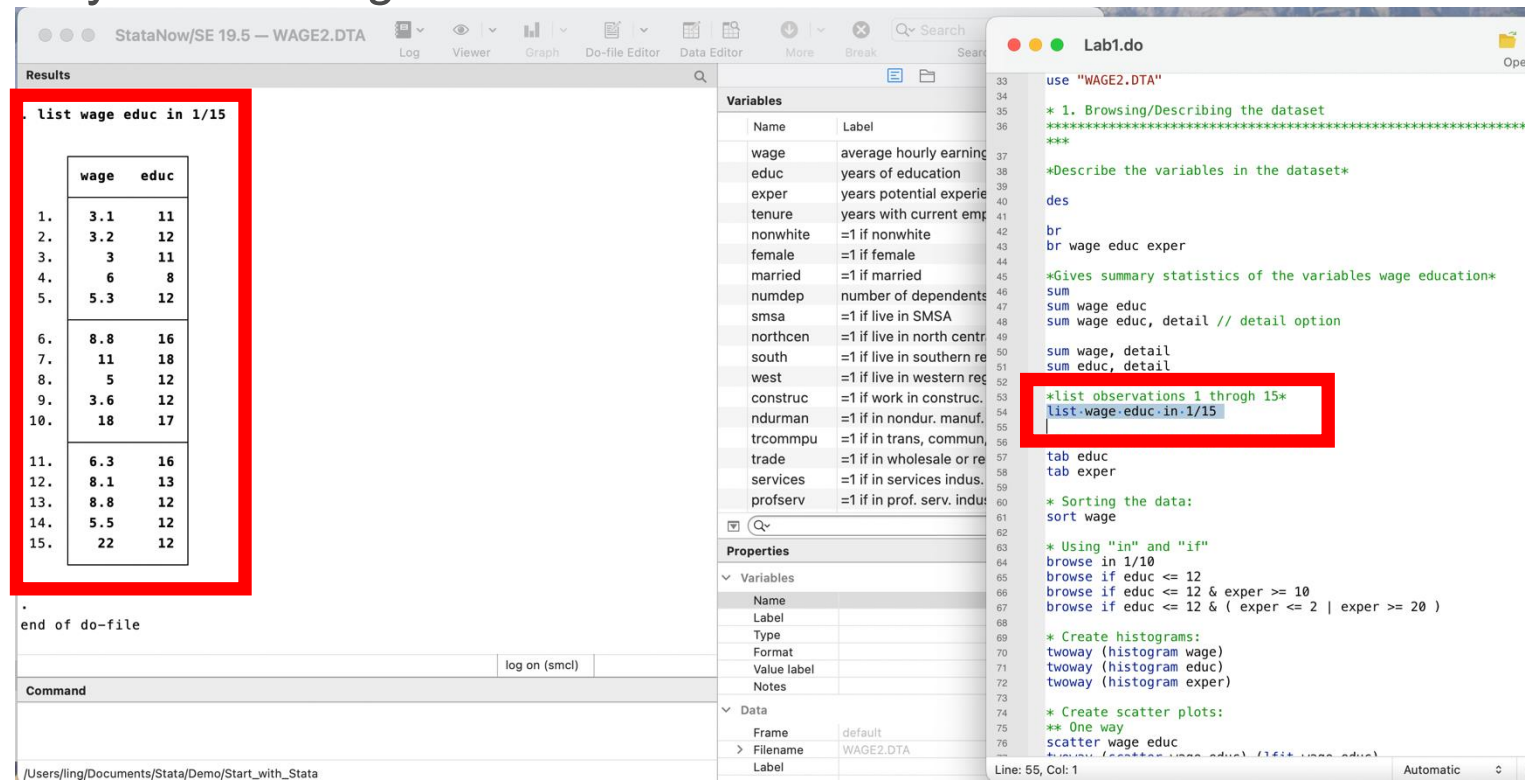
	wage	educ	exper
1	3.1	11	2
2	3.2	12	22
3	3	11	2
4	6	8	44
5	5.3	12	7
6	8.8	16	9
7	11	18	15
8	5	12	5
9	3.6	12	26
10	18	17	22
11	6.3	16	8
12	8.1	13	3
13	8.8	12	15
14	5.5	12	18
15	22	12	31
16	17	16	14
17	7.5	12	10
18	11	13	16
19	3.6	12	13
20	4.5	12	36
21	6.9	12	11
22	8.5	12	29
23	6.3	16	9
24	.53	12	3
25	6	11	37
26	9.6	16	3
27	7.8	16	11
28	13	16	31
29	13	15	30
30	3.3	8	9

Vars: 3 of 24 Order: Dataset Obs: 526

Filter: Off

List observations

- To list certain number of observations of columns we want to explore, we use the **list** command. The syntax is '**list variable name in row 1/row2**'. In this example, we list observations from row 1 through 15 and only look at *wage* and *educ* columns.



The screenshot shows the Stata software interface with the following components:

- Results Window:** Displays the output of the command `list wage educ in 1/15`. The output is a table with 15 rows and 2 columns: *wage* and *educ*.
- Command Window:** Shows the Stata script used to generate the results. The command `list wage educ in 1/15` is highlighted in a red box.
- Variables Window:** Lists the variables in the dataset, including *wage*, *educ*, *exper*, *tenure*, *nonwhite*, *female*, *married*, *numdep*, *smsa*, *northcen*, *south*, *west*, *construc*, *ndurman*, *trcommpu*, *trade*, *services*, and *profserv*.
- Properties Window:** Shows the properties of the variables, including Name, Label, Type, Format, Value label, and Notes.

	wage	educ
1.	3.1	11
2.	3.2	12
3.	3	11
4.	6	8
5.	5.3	12
6.	8.8	16
7.	11	18
8.	5	12
9.	3.6	12
10.	18	17
11.	6.3	16
12.	8.1	13
13.	8.8	12
14.	5.5	12
15.	22	12

```
use "WAGE2.DTA"
* 1. Browsing/Describing the dataset
*****
*Describe the variables in the dataset*
des
br
br wage educ exper
*Gives summary statistics of the variables wage education*
sum
sum wage educ
sum wage educ, detail // detail option
sum wage, detail
sum educ, detail
*list observations 1 through 15*
list wage educ in 1/15
tab educ
tab exper
* Sorting the data:
sort wage
* Using "in" and "if"
browse in 1/10
browse if educ <= 12
browse if educ <= 12 & exper >= 10
browse if educ <= 12 & ( exper <= 2 | exper >= 20 )
* Create histograms:
tsway (histogram wage)
tsway (histogram educ)
tsway (histogram exper)
* Create scatter plots:
** One way
scatter wage educ
* Two way (scatter wage educ) (ifit wage educ)
```

Summarize

- The **summarize/sum** command returns summary statistics, including the mean, standard deviation, minimum, and maximum values of each variable in the data set. For readability, Stata blocks off the data in groups of 5 variables.

The screenshot displays the Stata 19.5 interface with the 'WAGE2.DTA' dataset loaded. The 'Results' window shows the output of the `sum` command, which is highlighted with a red box. The output table provides summary statistics for 15 variables, grouped in blocks of 5. The 'Command' window at the bottom shows the command `. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"` being executed.

Variable	Obs	Mean	Std. dev.	Min	Max
wage	526	5.896103	3.693086	.53	24.98
educ	526	12.56274	2.769022	0	18
exper	526	17.01711	13.57216	1	51
tenure	526	5.104563	7.224462	0	44
nonwhite	526	.1026616	.3038053	0	1
female	526	.4790875	.500038	0	1
married	526	.608365	.4885804	0	1
numdep	526	1.043726	1.261891	0	6
smsa	526	.7224335	.4482246	0	1
northcen	526	.2509506	.4339728	0	1
south	526	.3555133	.4791242	0	1
west	526	.1692015	.3752867	0	1
construc	526	.0456274	.2088743	0	1
ndurman	526	.1140684	.318197	0	1
trcommpu	526	.0437262	.20468	0	1
trade	526	.2870722	.4528262	0	1
services	526	.1007605	.3012978	0	1
profserv	526	.2585551	.4382574	0	1

The 'Command' window shows the following command being executed:

```
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"
```

Summarize (cont'd)

- To see summary statistics for a subset of variables, use the syntax '*summarize/sum var 1 var 2 ...*'. Below, we print the summary statistics only for the variables *wage* and *educ*.

The screenshot displays the StataNow/SE 19.5 interface. The main window shows a do-file with the following commands:

```
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. br wage educ exper

. end of do-file

. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. br wage educ exper

. end of do-file

. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. sum wage educ

. end of do-file
```

The Results window shows the output of the `sum wage educ` command, which is highlighted with a red box:

Variable	Obs	Mean	Std. dev.	Min	Max
wage	526	5.896103	3.693086	.53	24.98
educ	526	12.56274	2.769022	0	18

The Variables window shows the list of variables in the dataset, including *wage*, *educ*, *exper*, *tenure*, *nonwhite*, *female*, *married*, *numdep*, *smsa*, *northcen*, *south*, *west*, *construc*, *ndurman*, *trcompu*, *trade*, *services*, and *profserv*.

The Command window shows the command `sum wage educ` being executed.

The Lab1.do window shows the following code:

```
use "WAGE2.DTA"

* 1. Browsing/Describing the dataset
*****
*Describe the variables in the dataset*
des
br
br wage educ exper

*Gives summary statistics of the variables wage educ
sum wage educ
sum wage, detail
sum educ, detail

*list observations 1 through 15*
list wage educ in 1/15

*Tabulate Distribution of Education *
tab educ
tab exper

* Sorting the data:
sort wage

* Using "in" and "if"
browse in 1/10
browse if educ <= 12
browse if educ <= 12 & exper >= 10
browse if educ <= 12 & ( exper <= 2 | exper >= 20 )

* Create histograms:
tsway (histogram wage)
tsway (histogram educ)
tsway (histogram exper)

* Create scatter plots:
** One way
scatter wage educ
tsway (scatter wage educ) (lfit wage educ)
```

Summarize (cont'd)

- We can use the **detail** option with the **summarize/sum** command. With **detail**, we get additional summary statistics for each variable, as well as values that correspond to the percentiles of the variable's distribution in the data. The 'Smallest' column reports the 4 smallest values in the data set and the 'Largest' column

The screenshot displays the StataNow/SE 19.5 interface. The Results window shows the output of the command `. sum wage educ, detail // detail option`. The output is divided into two sections: 'average hourly earnings' and 'years of education'. Each section provides a table of percentiles (1%, 5%, 10%, 25%, 50%, 75%, 90%, 95%, 99%) and the smallest and largest values. The Command window shows the command `. sum wage educ, detail // detail option`. The Variables window shows the list of variables and their labels.

Results

```
. sum wage educ, detail // detail option
```

average hourly earnings

Percentiles	Smallest		
1%	1.67	.53	
5%	2.75	1.43	
10%	2.92	1.5	Obs 526
25%	3.33	1.5	Sum of wgt. 526
50%	4.65		Mean 5.896103
75%	6.88	21.86	Std. dev. 3.693086
90%	10	22.2	Variance 13.63888
95%	13	22.86	Skewness 2.007325
99%	20	24.98	Kurtosis 7.970083

years of education

Percentiles	Smallest		
1%	4	0	
5%	8	0	
10%	9	2	Obs 526
25%	12	3	Sum of wgt. 526
50%	12		Mean 12.56274
		2.769022	Std. dev. 2.769022

log on (smcl)

Command

```
. sum wage educ, detail // detail option
```

Variables

Name	Label
wage	average hourly earning
educ	years of education
exper	years potential experie
tenure	years with current emp
nonwhite	=1 if nonwhite
female	=1 if female
married	=1 if married
numdep	number of dependents
smsa	=1 if live in SMSA
northcen	=1 if live in north centr
south	=1 if live in southern re
west	=1 if live in western reg
construc	=1 if work in construc.
ndurman	=1 if in nondur. manuf.
trcompu	=1 if in trans, commun
trade	=1 if in wholesale or re
services	=1 if in services indus.
profserv	=1 if in prof. serv. indu

Properties

Variables

Name	Label	Type	Format	Value label	Notes

Data

Frame	Filename	Label
default	WAGE2.DTA	

Lab1.do

```
use "WAGE2.DTA"

* 1. Browsing/Describing the dataset
*****
***

*Describe the variables in the dataset*
des

br
br wage educ exper

*Gives summary statistics of the variables wage education*
sum
sum wage educ
sum wage educ, detail // detail option

sum wage, detail
sum educ, detail

*list observations 1 through 15*
list wage educ in 1/15

*Tabulate Distribution of Education *
tab educ
tab exper

* Sorting the data:
sort wage

* Using "in" and "if"
browse in 1/10
browse if educ <= 12
browse if educ <= 12 & exper >= 10
browse if educ <= 12 & ( exper <= 2 | exper >= 20 )

* Create histograms:
twoway (histogram wage)
twoway (histogram educ)
twoway (histogram exper)

* Create scatter plots:
** One way
scatter wage educ
twoway (scatter wage educ) (1fit wage educ)
```

Line: 49, Col: 1

Tabulate

- The **tabulate/tab** command reports frequency counts for each value that a variable takes on in the data set. In the example below, there are 22 observations in the data set with *educ* = 8, they represent 4.18% of the data, and 7.60% of observations have an *educ* value of 8 or less.

The screenshot displays the Stata interface with the 'Results' window showing the output of the `tabulate educ` command. The output is a table with four columns: 'years of education', 'Freq.', 'Percent', and 'Cum.'. The table lists values from 0 to 18, with a total of 526 observations. The 'Percent' column shows the percentage of observations for each value, and the 'Cum.' column shows the cumulative percentage.

years of education	Freq.	Percent	Cum.
0	2	0.38	0.38
2	1	0.19	0.57
3	1	0.19	0.76
4	3	0.57	1.33
5	1	0.19	1.52
6	6	1.14	2.66
7	4	0.76	3.42
8	22	4.18	7.60
9	17	3.23	10.84
10	30	5.70	16.54
11	29	5.51	22.05
12	198	37.64	59.70
13	39	7.41	67.11
14	53	10.08	77.19
15	21	3.99	81.18
16	68	12.93	94.11
17	12	2.28	96.39
18	19	3.61	100.00
Total	526	100.00	

The 'Command' window shows the command `tabulate educ` being executed. The 'Variables' window shows the list of variables in the dataset, including *wage*, *educ*, *exper*, *tenure*, *nonwhite*, *female*, *married*, *numdep*, *smsa*, *northcen*, *south*, *west*, *construc*, *ndurman*, *trcommpu*, *trade*, *services*, and *profserv*.

Sort

- We use the **sort** command to arrange the observations in the dataset in ascending order, based on the values of a specific variable, such as sorting the data from the lowest to the highest *wage*.

The screenshot displays the Stata software interface with three main windows: Results, Variables, and a Do-file editor.

Results Window: Shows a summary of the dataset 'WAGE2.DTA'. The summary includes the following data:

5	1	0.19	1.52
6	6	1.14	2.66
7	4	0.76	3.42
8	22	4.18	7.60
9	17	3.23	10.84
10	30	5.70	16.54
11	29	5.51	22.05
12	198	37.64	59.70
13	39	7.41	67.11
14	53	10.08	77.19
15	21	3.99	81.18
16	68	12.93	94.11
17	12	2.28	96.39
18	19	3.61	100.00
Total	526	100.00	

Variables Window: Lists the variables in the dataset with their labels:

Name	Label
wage	average hourly earning
educ	years of education
exper	years potential experience
tenure	years with current employer
nonwhite	=1 if nonwhite
female	=1 if female
married	=1 if married
numdep	number of dependents
smsa	=1 if live in SMSA
northcen	=1 if live in north central
south	=1 if live in southern region
west	=1 if live in western region
construc	=1 if work in construction
ndurman	=1 if in nondurable manufacturing
trcommpu	=1 if in transportation, communication, and public utilities
trade	=1 if in wholesale and retail trade
services	=1 if in services industries
profserv	=1 if in professional, scientific, and technical services

Do-file Editor (Lab1.do): Contains Stata commands for data analysis. The command `sort wage` is highlighted with a red box. The do-file includes commands for summarizing data, creating histograms, and scatter plots.

```
sum
sum wage educ
sum wage educ, detail // detail option
sum wage, detail
sum educ, detail
*list observations 1 through 15*
list wage educ in 1/15
*Tabulate Distribution of Education *
tab educ
tab exper
* Sorting the data:
sort wage
* Using "in" and "if"
browse in 1/10
browse if educ <= 12
browse if educ <= 12 & exper >= 10
browse if educ <= 12 & ( exper <= 2 | exper >= 20 )
* Create histograms:
twoway (histogram wage)
twoway (histogram educ)
twoway (histogram exper)
* Create scatter plots:
** One way
scatter wage educ
twoway (scatter wage educ) (lfit wage educ)
** More interesting way
gen mean_wage_by_educ=.
forvalues i=0(1)18 {
sum wage if educ==`i'
replace mean_wage_by_educ=r(mean) if educ==`i'
}
scatter mean_wage_by_educ educ
twoway (scatter mean_wage_by_educ educ) (lfit mean_wage_by_educ educ)
* Create bar graph:
graph bar (mean) wage, over(educ)
```

Generate

- To generate a new variable in the data set, we use the **generate/gen** command. For example, **gen educsq = educ^2** creates a new column containing the squared value of education for each person, which is often used in labor economics to model how the “returns” to schooling might change over time. Also note that our new variable has populated in the variables panel, but is missing a label

Variables	
Name	Label
numdep	number of dependents
smsa	=1 if live in SMSA
northcen	=1 if live in north central
south	=1 if live in southern region
west	=1 if live in western region
construc	=1 if work in construction
ndurman	=1 if in nondurable manufacturing
trcommpu	=1 if in transportation, communication, and public utilities
trade	=1 if in wholesale and retail trade
services	=1 if in services industries
profserv	=1 if in professional, scientific, and technical services
profocc	=1 if in professional, scientific, and technical occupations
clerocc	=1 if in clerical, sales, and related occupations
servocc	=1 if in service occupations
lwage	log(wage)
expersq	exper^2
tenursq	tenure^2
educsq	

```
73
74 * Create scatter plots:
75 ** One way
76 scatter wage educ
77 twoway (scatter wage educ) (lfit wage educ)
78
79 ** More interesting way
80 gen mean_wage_by_educ=.
81 forvalues i=0(1)18 {
82   sum wage if educ==`i'
83   replace mean_wage_by_educ=r(mean) if educ==`i'
84 }
85
86 scatter mean_wage_by_educ educ
87 twoway (scatter mean_wage_by_educ educ) (lfit mean
88   educ)
89
90 * Create bar graph:
91 graph bar (mean) wage, over(educ)
92
93 * Create new variables:
94 gen educsq = educ^2
95
96 tab indicator
97
98 * Renaming variables:
99 rename indicator highschool
100
101 * Labeling variables:
102 label var highschool "1 if worker completed high school"
```

Generate (cont'd)

- We also use **gen** to create "dummy" or indicator variables that take the value of 1 if a condition is true and 0 otherwise. For example, `gen indicator = (educ >= 12)` creates a new variable that equals 1 for everyone who has at least a high school education and 0 for everyone else, allowing us to easily compare these two groups

Variables		
Name	Label	
smsa	=1 if live in SMSA	
northcen	=1 if live in north central	
south	=1 if live in southern region	
west	=1 if live in western region	
construc	=1 if work in construction	
ndurman	=1 if in nondurable manufacturing	
trcommpu	=1 if in transportation, communication, and public utilities	
trade	=1 if in wholesale and retail trade	
services	=1 if in services industries	
profserv	=1 if in professional, scientific, and technical services	
profocc	=1 if in professional, scientific, and technical occupations	
clerocc	=1 if in clerical, sales, and related occupations	
servocc	=1 if in service occupations	
lwage	log(wage)	
expersq	exper ²	
tenursq	tenure ²	
educsq		
indicator		

```
74 * Create scatter plots:
75 ** One way
76 scatter wage educ
77 twoway (scatter wage educ) (lfit wage educ)
78
79 ** More interesting way
80 gen mean_wage_by_educ=.
81 forvalues i=0(1)18 {
82   sum wage if educ==`i'
83   replace mean_wage_by_educ=r(mean) if educ==`i'
84 }
85
86 scatter mean_wage_by_educ educ
87 twoway (scatter mean_wage_by_educ educ)
88
89 * Create bar graph:
90 graph bar (mean) wage, over(educ)
91
92 * Create new variables:
93 gen educsq = educ^2
94 gen indicator = (educ >= 12)
95 label indicator
96
97 * Renaming variables:
98 rename indicator highschool
99
100 * Labeling variables:
101 label var highschool "1 = High School or more"
102
```

Drop

- We use the **drop** command to permanently remove variables from the current dataset that are no longer needed for our analysis. For example, typing **drop highschool** will delete that specific column, helping us keep our workspace clean and focused on only the variables relevant to our research question.

The screenshot shows the StataNow/SE 19.5 interface with the following components:

- Results Window:** Displays a summary table with 4 columns. The last row shows a total of 410 observations and a mean of 100.00.
- Variables Window:** Lists variables including smsa, northcen, south, west, construc, ndurman, trcommpu, trade, services, profserv, profocc, clerocc, servocc, lwage, expersq, tenursq, educsq, and mean...educ.
- Lab1.do File:** Contains Stata commands. The command `drop highschool` is highlighted in a red box.

```
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"  
  
. drop highschool  
  
. end of do-file
```

18	1	0.24	96.83
18	1	0.24	97.07
18	1	0.24	97.32
18	2	0.49	97.80
18	1	0.24	98.05
19	1	0.24	98.29
20	1	0.24	98.54
20	1	0.24	98.78
22	1	0.24	99.02
22	1	0.24	99.27
22	1	0.24	99.51
23	1	0.24	99.76
25	1	0.24	100.00
Total	410	100.00	

Variables

Name	Label
smsa	=1 if live in SMSA
northcen	=1 if live in north central U.S.
south	=1 if live in southern region
west	=1 if live in western region
construc	=1 if work in construc. indus.
ndurman	=1 if in nondur. manuf. indus.
trcommpu	=1 if in trans, commun, pub ut
trade	=1 if in wholesale or retail
services	=1 if in services indus.
profserv	=1 if in prof. serv. indus.
profocc	=1 if in profess. occupation
clerocc	=1 if in clerical occupation
servocc	=1 if in service occupation
lwage	log(wage)
expersq	exper^2
tenursq	tenure^2
educsq	
mean...educ	

Properties

Name	Label

```
80 gen mean_wage_by_educ=.  
81 forvalues i=0(1)18 {  
82   sum wage if educ==`i'  
83   replace mean_wage_by_educ=r(mean)  
84 }  
85  
86 scatter mean_wage_by_educ educ  
87 twoway (scatter mean_wage_by_educ  
88   educ)  
89  
90 * Create bar graph:  
91 graph bar (mean) wage, over(educ)  
92  
93 * Create new variables:  
94 gen educsq = educ^2  
95 gen indicator = (educ>=12)  
96 tab indicator  
97  
98 * Renaming variables:  
99 rename indicator highschool  
100  
101 * Labeling variables:  
102 label var highschool "1 if worker"  
103  
104 * Bysort:  
105 bysort highschool: tab wage  
106 bysort highschool: sum wage  
107  
108 * Delete variable  
109 drop highschool  
110  
111 * Compute mean:  
112 mean wage  
113 mean educ  
114 mean exper  
115  
116 * Compute correlations:
```

Bysort

- We use the **bysort** prefix to repeat a command for different groups within our data. For example, **bysort highschool: sum wage** tells Stata to first group the workers by their education level (high school graduates vs. non-graduates) and then calculate the average wage for each group separately, allowing us to see the "pay gap" between them immediately.

The screenshot shows the Stata interface with the command window on the left, the Results window in the center, and the Variables window on the right. The command window contains the following code:

```
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. bysort highschool: tab wage

-> highschool = 0
```

The Results window displays the output of the `bysort highschool: tab wage` command, showing a table of average hourly earnings by education level (highschool = 0). The table is highlighted with a red border.

average hourly earnings	Freq.	Percent	Cum.
1.5	1	0.86	0.86
1.7	1	0.86	1.72
1.8	1	0.86	2.59
2	1	0.86	3.45
2	2	1.72	5.17
2.2	1	0.86	6.03
2.3	1	0.86	6.90
2.3	1	0.86	7.76
2.3	1	0.86	8.62
2.4	1	0.86	9.48
2.5	1	0.86	10.34
2.6	1	0.86	11.21
2.7	1	0.86	12.07
2.8	1	0.86	12.93
2.9	1	0.86	13.79
2.9	2	1.72	15.52

The Variables window on the right lists the variables in the dataset, including `highschool` (1 if worker completed high school or more). The command window on the right shows the code for the `bysort` command, which is highlighted with a red border:

```
* Bysort:
bysort highschool: tab wage
bysort highschool: sum wage
```

If statements

- We use the **if** qualifier to restrict a command to observations that meet specific logical conditions, regardless of their row number. For example, **browse if educ <= 12** allows us to view only the individuals in the dataset who have 12 years of schooling or less.

The screenshot shows the StataNow/SE 19.5 interface. On the left, the 'Results' window displays the output of a do-file. The do-file contains the following commands:

```
.  
end of do-file  
Total 526 100.00  
.  
do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"  
.  
sort wage  
.  
end of do-file  
do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"  
.  
browse in 1/10  
.  
end of do-file  
do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"  
browse if educ <= 12  
.  
end of do-file  
.  
log on (smcl)  
Command
```

The 'Data Editor (Browse) — WAGE2.DTA' window is open, showing a subset of data where `educ <= 12`. The table has columns: wage, educ, exper, tenure, nonwhite, female, married, numdep, smsa, northcen, south, w. The first 34 rows are displayed, with the first row having wage = .53 and educ = 12.

On the right, the 'Variables' window shows a list of variables with their labels. The 'Properties' window for the 'wage' variable is also visible, showing its label as 'average hourly earnings', type as 'float', and format as '%8.2g'.

If statements (cont'd)

- **browse if educ <= 12 & exper >= 10**. This command filters the data to show only workers who have a high school education (or less) **and** at least 10 years of experience.

The screenshot shows the Stata 19.5 interface. On the left, the 'Results' window displays a do-file with the following commands:

```
. sort wage
*
end of do-file

. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. browse in 1/10

*
end of do-file

. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. browse if educ <= 12

*
end of do-file

. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. browse if educ <= 12 & exper >= 10

*
end of do-file
```

The 'Command' window at the bottom shows the command `log on (smcl)`.

The 'Data Editor (Browse) — WAGE2.DTA' window displays a dataset with the following variables: wage, educ, exper, tenure, nonwhite, female, married, numdep, smsa, northcen, south, and west. The 'wage' variable is selected, and the value 1.4 is shown in the top right corner. The dataset is filtered to show only workers with high school education or less (educ <= 12) and at least 10 years of experience (exper >= 10).

The 'Variables' list on the right shows the following variables and their labels:

- wage: average hou...
- educ: years of edu...
- exper: years potent...
- tenure: years with c...
- nonwhite: =1 if nonwhite
- female: =1 if female
- married: =1 if married
- numdep: number of d...
- smsa: =1 if live in S...
- northcen: =1 if live in n...
- south: =1 if live in s...
- west: =1 if live in...
- construc: =1 if work in...
- ndurman: =1 if in nond...
- trcommu: =1 if in trans...

The 'Properties' window on the right shows the properties for the selected variable 'wage':

- Name: wage
- Label: average hourly earnings
- Type: float
- Format: %8.2g
- Value label:
- Notes:

The 'Data' window on the right shows the default frame and the filename 'WAGE2.DTA'.

If statements (cont'd)

- ***browse if educ <= 12 & (exper <= 2 | exper >= 20)***. This allows us to find workers with low education who are either very new to the workforce (2 years or less) **or** are highly experienced veterans (20 years or more).

The screenshot displays the Stata software interface. On the left, the 'Results' window shows the execution of a do-file with the following commands:

```
. browse in 1/10
*
end of do-file

. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. browse if educ <= 12
*
end of do-file

. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. browse if educ <= 12 & exper >= 10
*
end of do-file

. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

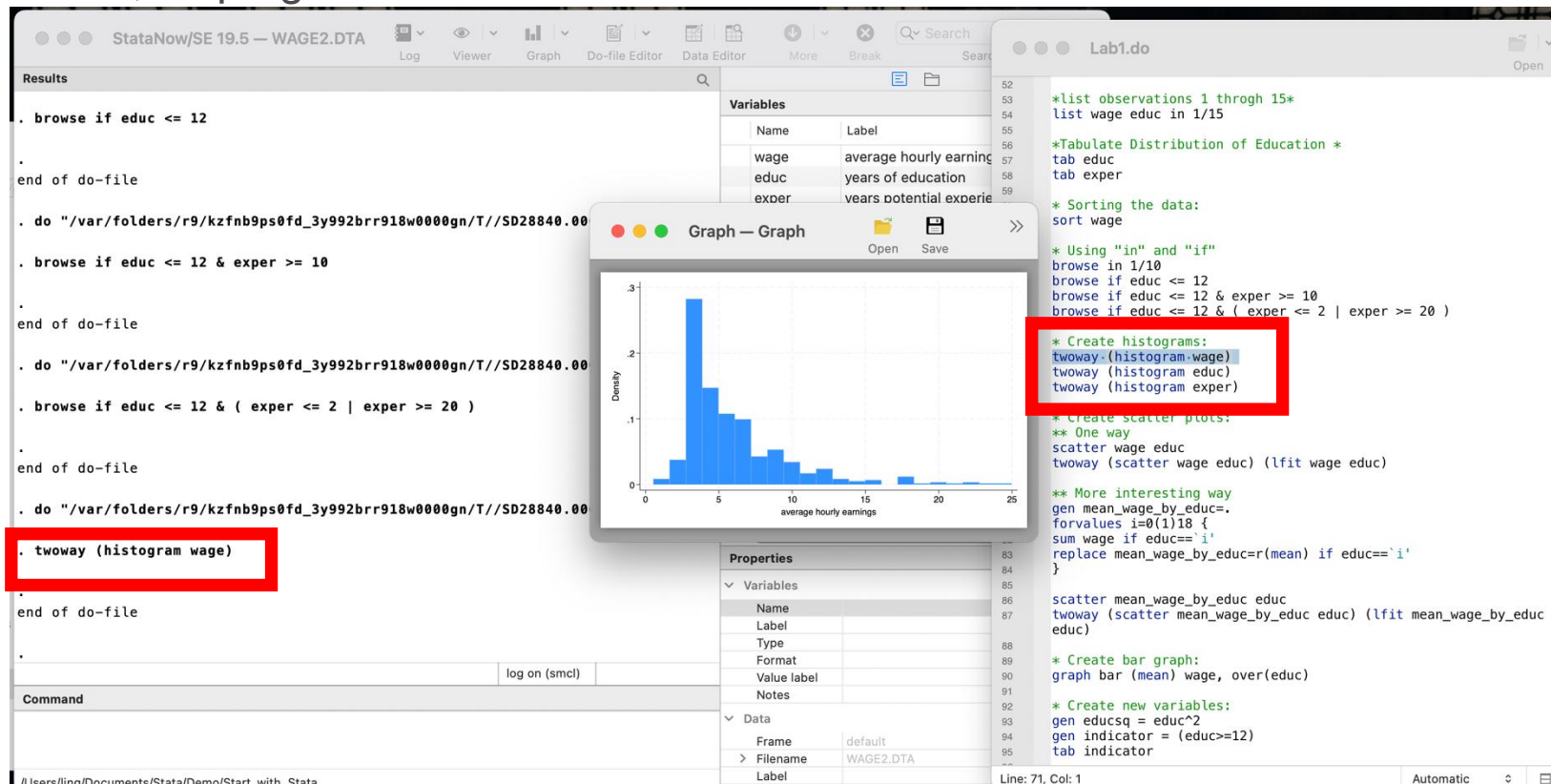
. browse if educ <= 12 & ( exper <= 2 | exper >= 20 )
*
end of do-file
```

The 'Command' window at the bottom shows the command `log on (smcl)`. The main window is the 'Data Editor (Browse) — WAGE2.DTA', showing a dataset with variables: wage, educ, exper, tenure, nonwhite, female, married, numdep, smsa, northcen, south, and w. The 'wage' variable is highlighted, showing a value of 1.5 for the first observation. The 'Variables' list on the right includes: wage (average hou...), educ (years of edu...), exper (years potent...), tenure (years with c...), nonwhite (=1 if nonwhite), female (=1 if female), married (=1 if married), numdep (number of d...), smsa (=1 if live in S...), northcen (=1 if live in n...), south (=1 if live in s...), west (=1 if live in...), construc (=1 if work in...), ndurman (=1 if in nond...), and trcommpu (=1 if in trans...).

	wage	educ	exper	tenure	nonwhite	female	married	numdep	smsa	northcen	south	w
4	1.5	8	31	30	0	0	0	0	0	0	1	
6	1.7	10	1	0	0	0	0	1	1	0	0	
7	1.8	11	2	1	0	0	0	2	1	0	0	
8	2	11	38	3	1	1	0	0	0	0	1	
11	2	9	2	0	0	0	0	3	0	1	0	
15	2.2	8	28	3	0	0	1	4	0	0	1	
17	2.3	10	2	0	0	1	0	3	0	0	0	
18	2.3	10	2	0	0	1	0	4	1	1	0	
21	2.5	12	2	0	0	0	0	2	0	0	0	
22	2.5	12	46	0	0	1	0	0	1	0	0	
23	2.5	9	2	0	0	1	0	1	1	0	1	
25	2.7	12	42	10	0	1	1	0	1	0	0	
26	2.7	9	2	0	0	1	0	1	1	1	0	
28	2.8	10	1	0	0	0	0	4	1	1	0	
29	2.9	12	45	8	0	1	1	0	0	1	0	
30	2.9	10	2	0	0	1	0	3	1	0	0	
31	2.9	0	42	0	0	1	1	2	0	0	0	
35	2.9	8	49	6	0	1	0	1	0	0	0	
36	2.9	12	39	1	0	1	1	0	0	0	1	
37	2.9	5	34	0	0	1	1	5	0	0	0	
38	2.9	12	26	1	0	1	0	4	1	0	1	
40	2.9	11	2	0	0	1	0	2	1	0	1	
41	2.9	11	1	0	0	1	0	1	1	1	0	
42	2.9	10	23	2	0	1	0	2	0	0	1	
43	2.9	6	49	7	0	0	1	0	0	0	1	
44	2.9	11	1	2	1	0	0	2	0	0	1	
46	2.9	12	37	0	0	1	1	0	0	0	1	
47	2.9	12	2	2	0	1	0	0	1	0	0	
50	2.9	11	1	0	0	1	0	3	1	1	0	
51	2.9	12	20	34	0	1	1	2	1	0	0	

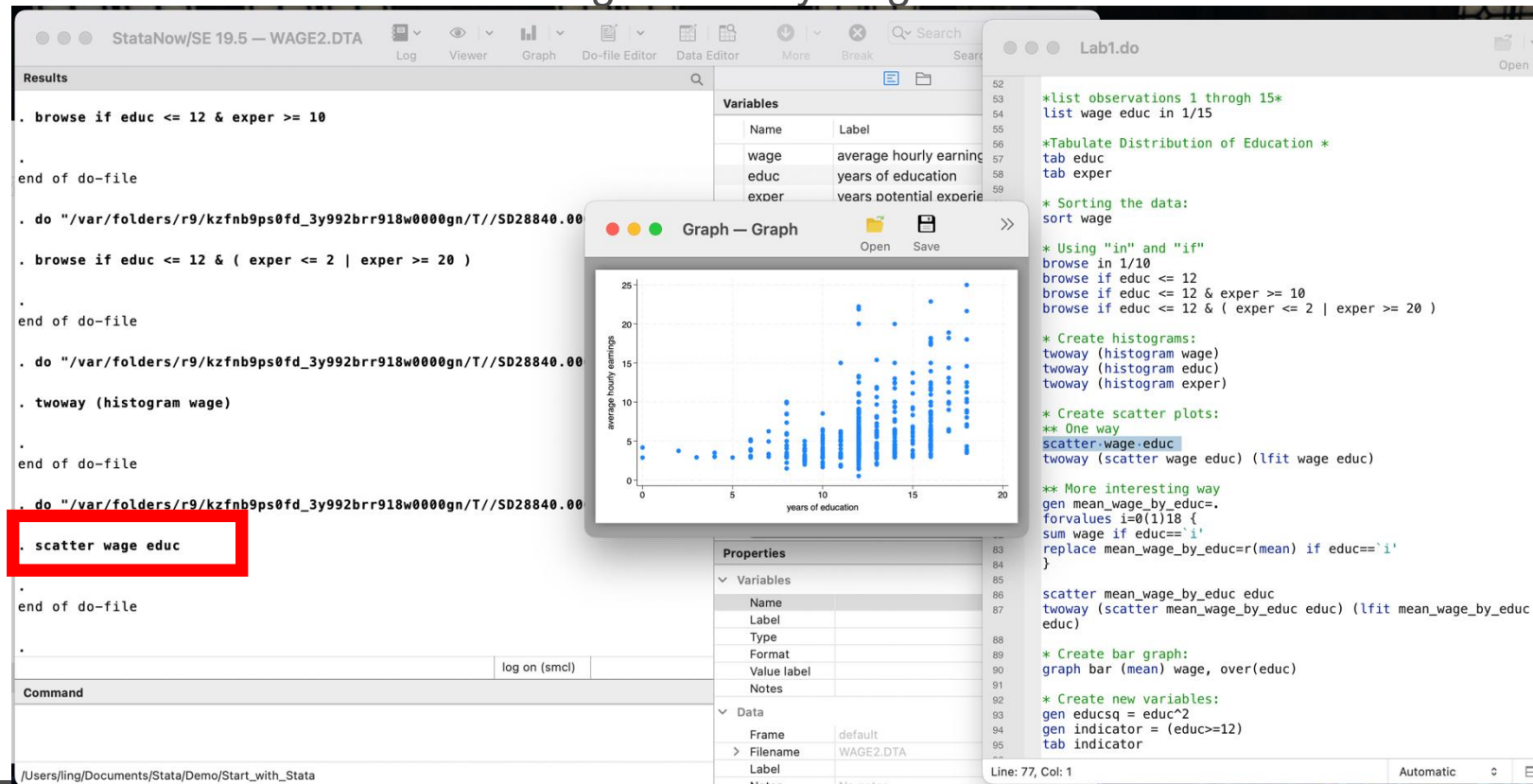
Figures: bar plot

- We use the **histogram** command to visualize the distribution of a single variable by grouping values into bins. For example, **twoway (histogram wage)** creates a chart that shows how frequently different wage levels occur in our dataset, helping us see if most workers earn a similar amount or if there is a wide spread.



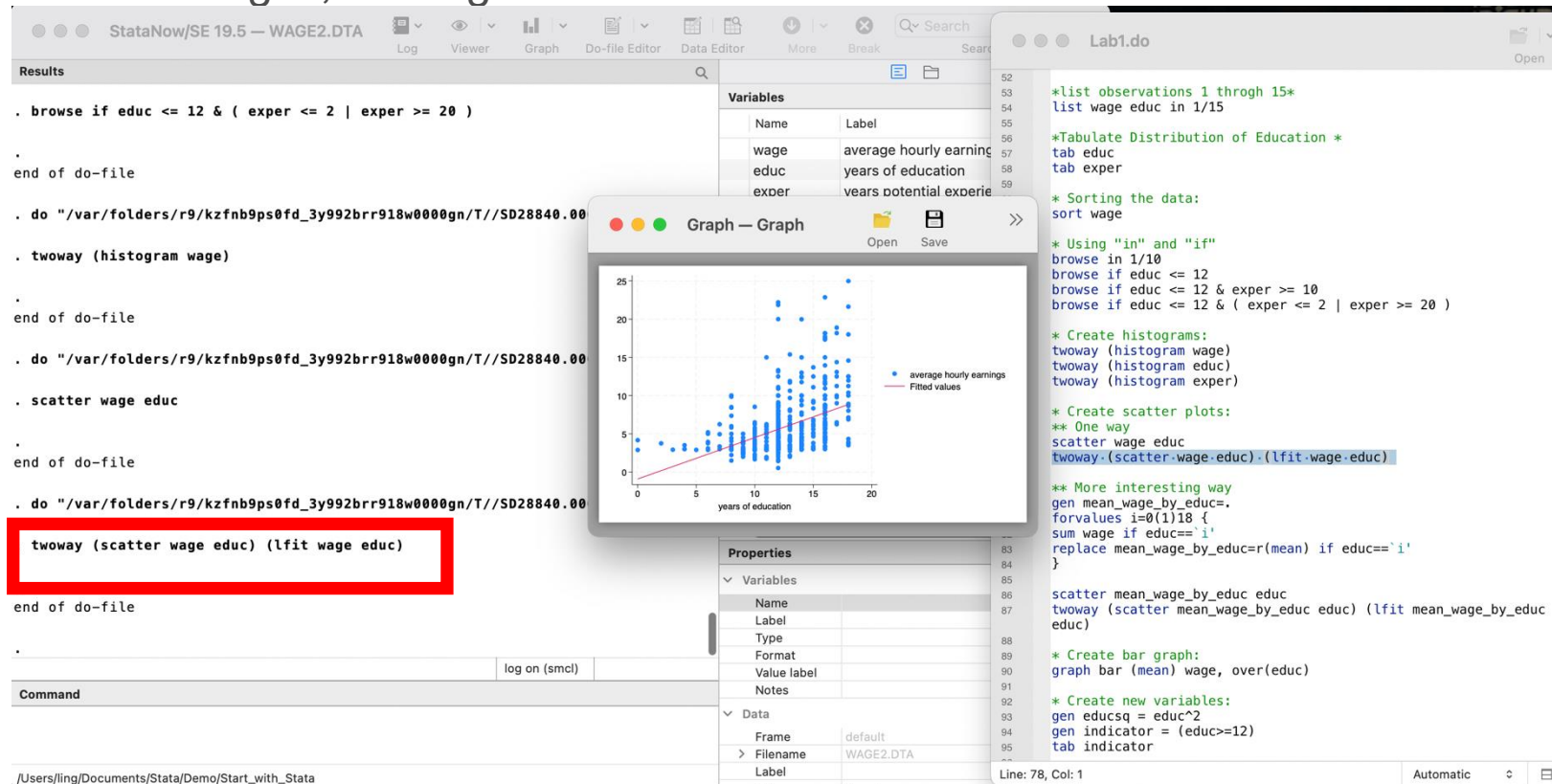
Figures: scatter plot

- We use the **scatter** command to explore the relationship between two different variables. For example, **scatter wage educ** creates a plot where each dot represents an individual worker, allowing us to see if those with more years of education tend to earn higher hourly wages.



Figures: scatter plot (cont'd)

- To better see the overall trend in a scatter plot, we can use **twoway (scatter wage educ) (lfit wage educ)** to overlay a "line of best fit" on top of our data points. This straight line represents the average relationship between education and wages, making it easier to see the correlation between the two.



In

- We use the *in* qualifier to restrict a command to a specific range of rows based on their current order in the dataset. For example, *browse in 1/10* opens the Data Editor to show only the first ten observations.

The screenshot displays the Stata software interface. On the left, the 'Results' window shows the output of a do-file. The do-file contains the following commands:

```
*  
end of do-file  
  
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"  
  
. sort wage  
  
*  
end of do-file  
  
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"  
  
. browse in 1/10  
  
*  
end of do-file  
  
*  
log on (smcl)  
  
Command
```

The 'browse in 1/10' command is highlighted with a red rectangle. On the right, the 'Data Editor (Browse) — WAGE2.DTA' window is open, showing the first 10 observations of the dataset. The variables displayed are wage, educ, exper, tenure, nonwhite, female, married, numdep, smsa, northcen, south, and west. The 'wage' variable is selected in the 'Variables' list on the right. The 'Properties' window on the right shows the details for the 'wage' variable, including its label 'average hourly earnings', type 'float', and format '%8.2g'.

Rename

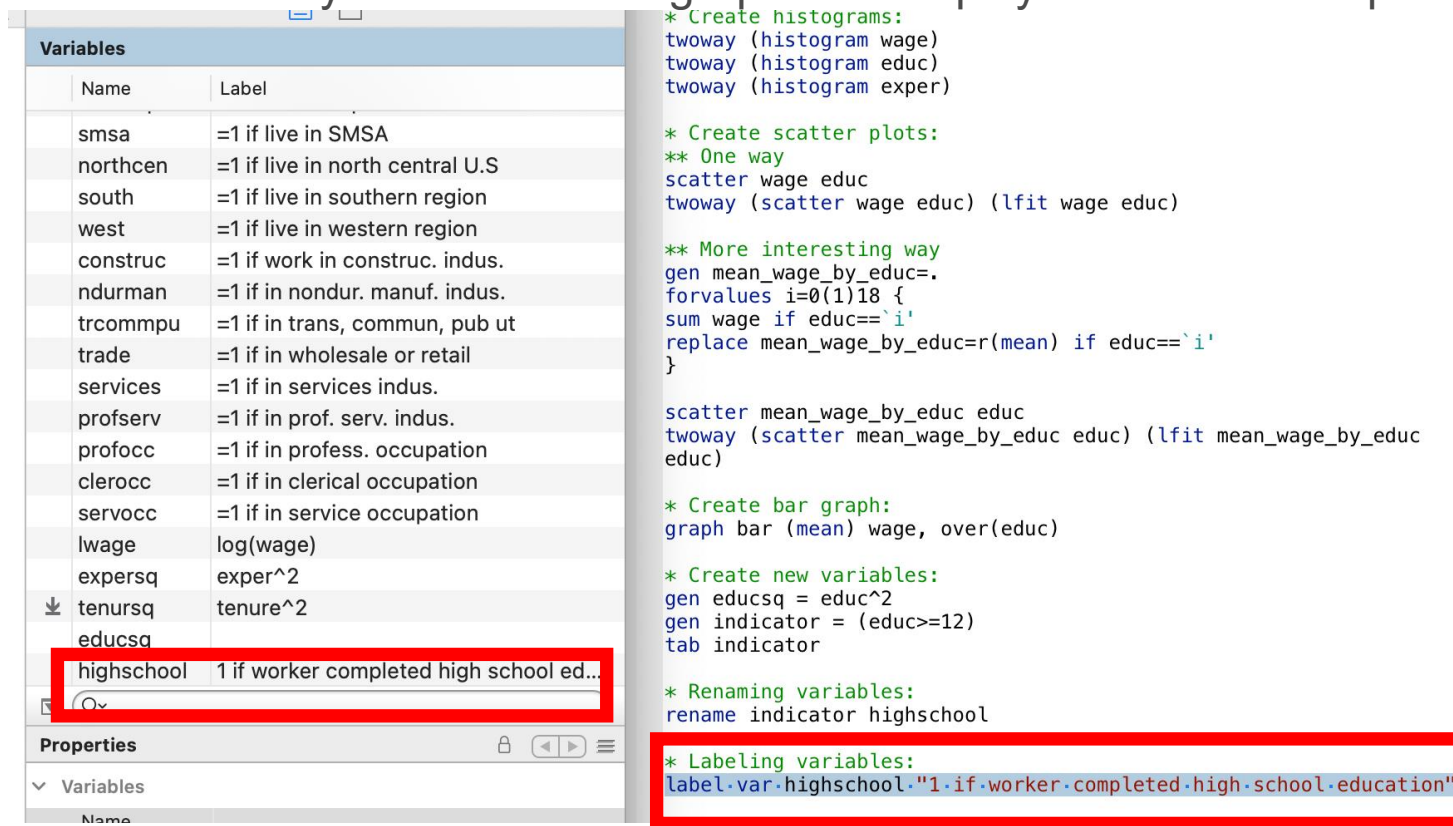
- We use the **rename** command to give an existing variable a more descriptive or convenient name. For example, if we decide that "indicator" is too vague, we can use **rename indicator highschool** to change the column header, making our dataset and future command results much easier to read and understand..

Variables	
Name	Label
smsa	=1 if live in SMSA
northcen	=1 if live in north centr
south	=1 if live in southern re
west	=1 if live in western reg
construc	=1 if work in construc.
ndurman	=1 if in nondur. manuf.
trcommpu	=1 if in trans, commun,
trade	=1 if in wholesale or re
services	=1 if in services indus.
profserv	=1 if in prof. serv. indus
profocc	=1 if in profess. occupa
clerocc	=1 if in clerical occupa
servocc	=1 if in service occupa
lwage	log(wage)
expersq	exper^2
tenursq	tenure^2
educsq	
highschool	

```
73
74 * Create scatter plots:
75 ** One way
76 scatter wage educ
77 twoway (scatter wage educ) (lfit wage educ)
78
79 ** More interesting way
80 gen mean_wage_by_educ=.
81 forvalues i=0(1)18 {
82   sum wage if educ==`i'
83   replace mean_wage_by_educ=r(mean) if educ==`i'
84 }
85
86 scatter mean_wage_by_educ educ
87 twoway (scatter mean_wage_by_educ educ) (lfit mean_wa
educ)
88
89 * Create bar graph:
90 graph bar (mean) wage, over(educ)
91
92 * Create new variables:
93 gen educsq = educ^2
94 gen indicator = (educ>=12)
95 tab indicator
96
97 * Renaming variables:
98 rename indicator highschool
99
100 * Labeling variables:
101 label var highschool "1 if worker completed high sch
```

Label

- We use the **label var** command to attach a descriptive definition to a variable name, which serves as a helpful reminder of what the data represents. For example, by typing **label var highschool "1 if worker completed high school education"**, we ensure that any future tables or graphs will display this clear description instead of just the short variable name.



The screenshot displays the Stata interface. On the left, the 'Variables' list shows a table of variables and their labels. The 'highschool' variable is highlighted with a red box. On the right, the Do-file editor shows Stata commands. The 'label var' command is highlighted with a red box.

Name	Label
smsa	=1 if live in SMSA
northcen	=1 if live in north central U.S
south	=1 if live in southern region
west	=1 if live in western region
construc	=1 if work in construc. indus.
ndurman	=1 if in nondur. manuf. indus.
trcommpu	=1 if in trans, commun, pub ut
trade	=1 if in wholesale or retail
services	=1 if in services indus.
profserv	=1 if in prof. serv. indus.
profocc	=1 if in profess. occupation
clerocc	=1 if in clerical occupation
servocc	=1 if in service occupation
lwage	log(wage)
expersq	exper^2
tenursq	tenure^2
educsq	
highschool	1 if worker completed high school ed...

```
* Create histograms:
tway (histogram wage)
tway (histogram educ)
tway (histogram exper)

* Create scatter plots:
** One way
scatter wage educ
tway (scatter wage educ) (lfit wage educ)

** More interesting way
gen mean_wage_by_educ=.
forvalues i=0(1)18 {
sum wage if educ==`i'
replace mean_wage_by_educ=r(mean) if educ==`i'
}

scatter mean_wage_by_educ educ
tway (scatter mean_wage_by_educ educ) (lfit mean_wage_by_educ
educ)

* Create bar graph:
graph bar (mean) wage, over(educ)

* Create new variables:
gen educsq = educ^2
gen indicator = (educ>=12)
tab indicator

* Renaming variables:
rename indicator highschool

* Labeling variables:
label var highschool "1 if worker completed high school education"
```

More interesting way...

- We use a **forvalues loop** to automate the process of calculating group-specific statistics. In this example, the code tells Stata to look at every education level from 0 to 18 years, calculate the average wage for that specific group, and save that average into a new variable called **mean_wage_by_educ**. This "condenses" hundreds of individual workers into a single representative average for each level of schooling.

The screenshot displays the Stata interface with the following components:

- Command Window:** Contains the following code:

```
gen mean_wage_by_educ=.
526 missing values generated)

forvalues i=0(1)18 {
  2. sum wage if educ==`i'
  3. replace mean_wage_by_educ=r(mean) if educ==`i'
  4. }
```
- Results Window:** Shows the output of the command, including summary statistics for the wage variable across different education levels. The output is as follows:

Variable	Obs	Mean	Std. dev.	Min	Max
wage	2	3.53	.9050967	2.89	4.17
2 real changes made)					
Variable	Obs	Mean	Std. dev.	Min	Max
wage	0				
0 real changes made)					
Variable	Obs	Mean	Std. dev.	Min	Max
wage	1	3.75	.	3.75	3.75
1 real change made)					
Variable	Obs	Mean	Std. dev.	Min	Max
wage	1	2.92	.	2.92	2.92
1 real change made)					
- Lab1.do Editor:** Contains the following code:

```
* Create histograms:
twoway (histogram wage)
twoway (histogram educ)
twoway (histogram exper)

* Create scatter plots:
** One way
scatter wage educ
twoway (scatter wage educ) (lfit wage educ)

** More interesting way
gen mean_wage_by_educ=.
forvalues i=0(1)18 {
  sum wage if educ==`i'
  replace mean_wage_by_educ=r(mean) if educ==`i'
}

scatter mean_wage_by_educ educ
twoway (scatter mean_wage_by_educ educ) (lfit mean_wage_by_educ educ)

* Create bar graph:
graph bar (mean) wage, over(educ)

* Create new variables:
gen educsq = educ^2
gen indicator = (educ>=12)
tab indicator

* Renaming variables:
rename indicator highschool

* Labeling variables:
label var highschool "1 if worker completed high school education"

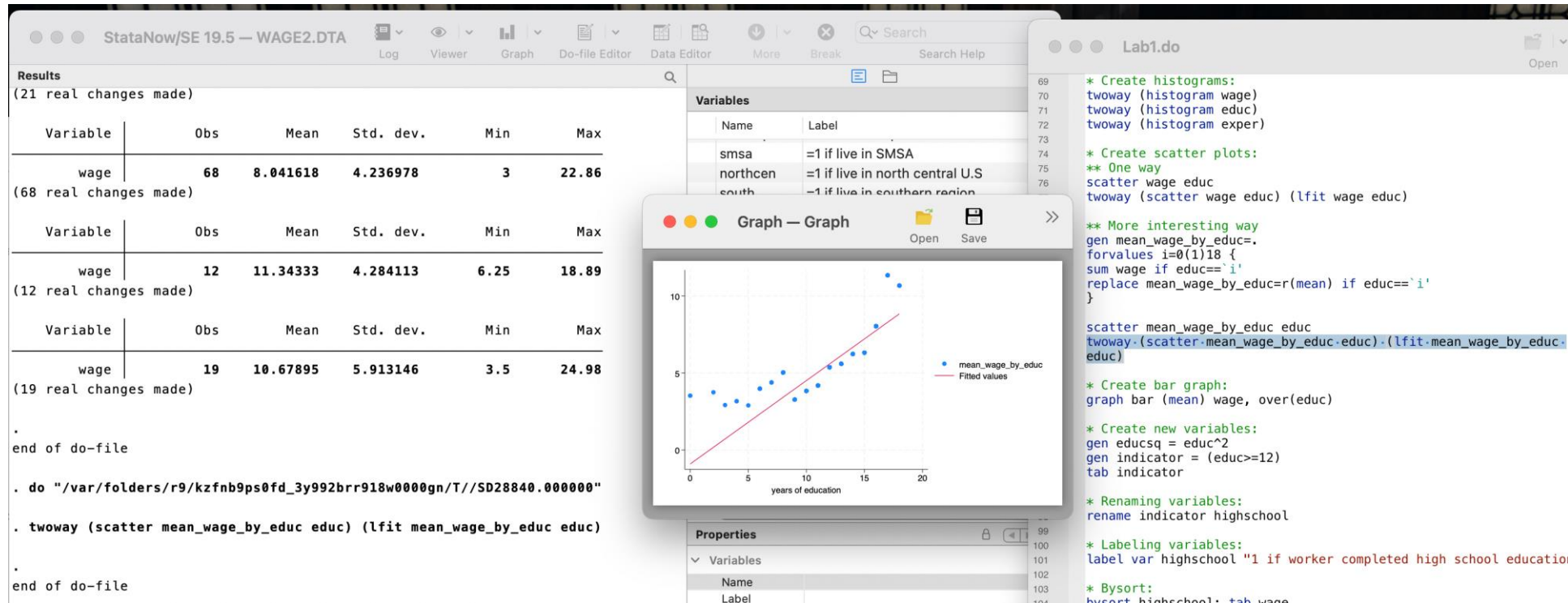
* Bysort:
bysort highschool: tab wage
bysort highschool: sum wage

* Delete variable
drop highschool

* Compute mean:
mean wage
mean educ
```

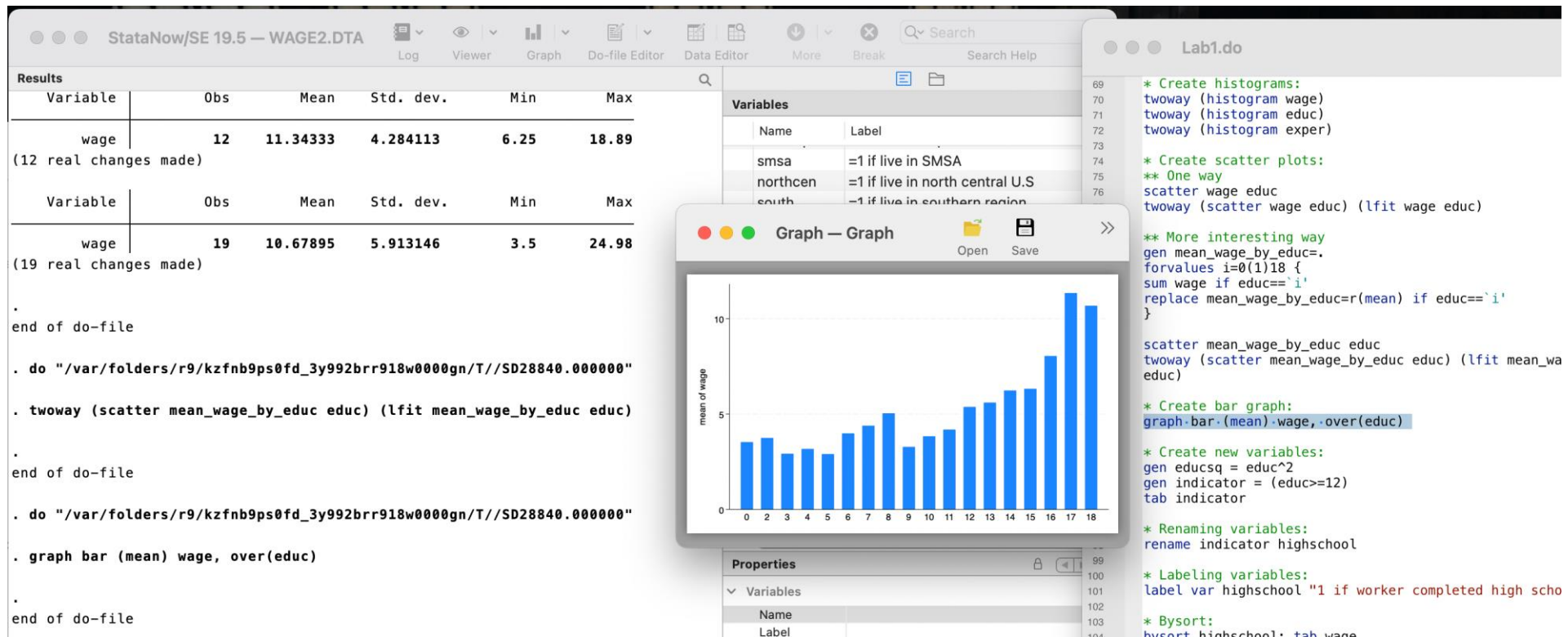
More interesting way (cont'd)

- Once we have these averages, we use ***twoway (scatter mean_wage_by_educ educ) (lfit mean_wage_by_educ educ)*** to visualize the trend. Unlike a standard scatter plot that shows every worker as a dot, this version plots only the "mean" points, making it much easier to see the clear, upward slope that represents how earnings typically rise with more education.



More interesting way (cont'd)

- We use the **graph bar (mean)** command to create a visual comparison of averages across categories. By using the **over(educ)** option, Stata generates a series of bars where the height of each bar represents the average hourly wage for that specific education level, providing a clear "at-a-glance" view of the pay differences between groups.



Mean

- We use the **mean** command to calculate the average value of a variable along with its standard error and confidence interval. Unlike the sum command, which provides a quick snapshot, mean wage treats the data as a sample and provides statistical estimates that help us understand how precisely we are measuring the average earnings of the population.

The screenshot displays the StataNow/SE 19.5 interface. The main window shows the command log with the following commands:

```
. end of do-file  
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"  
  
. drop highschool  
  
. end of do-file  
  
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"  
  
. mean wage
```

The output of the `. mean wage` command is shown in a table, which is highlighted with a red border:

Mean estimation		Number of obs = 526		
	Mean	Std. err.	[95% conf. interval]	
wage	5.896103	.1610262	5.579768	6.212437

The right-hand pane shows the list of variables in the dataset:

Name	Label
smsa	=1 if live in SMSA
northcen	=1 if live in north central U.S.
south	=1 if live in southern region
west	=1 if live in western region
construc	=1 if work in construc. indus.
ndurman	=1 if in nondur. manuf. indus.
trcommpu	=1 if in trans, commun, pub ut
trade	=1 if in wholesale or retail
services	=1 if in services indus.
profserv	=1 if in prof. serv. indus.
profocc	=1 if in profess. occupation
clerocc	=1 if in clerical occupation
servocc	=1 if in service occupation
lwage	log(wage)
expersq	exper^2
tenursq	tenure^2
educsq	
mean...educ	

The bottom-right pane shows the command window with the following commands:

```
80 gen mean_wage_by_educ  
81 forvalues i=0(1)18 {  
82 sum wage if educ==`i'  
83 replace mean_wage_by_educ,`i'  
84 }  
85  
86 scatter mean_wage_by_educ,`i'  
87 twoway (scatter mean_wage_by_educ,`i')  
88  
89 * Create bar graph:  
90 graph bar (mean) wage,`i'  
91  
92 * Create new variable:  
93 gen educsq = educ^2  
94 gen indicator = (educ>18)  
95 tab indicator  
96  
97 * Renaming variables:  
98 rename indicator high  
99  
100 * Labeling variables:  
101 label var highschool `i'  
102  
103 * Bysort:  
104 bysort highschool: tal  
105 bysort highschool: sur  
106  
107 * Delete variable  
108 drop highschool  
109  
110 * Compute mean:  
111 mean wage  
112 mean educ  
113 mean exper  
114
```

Correlations

- We use the **corr** command to measure the strength and direction of the linear relationship between two or more variables.
 - **Multiple Variables:** Typing **corr wage educ exper tenure** produces a "correlation matrix" that shows how every variable relates to every other variable simultaneously.

The screenshot displays the StataNow/SE 19.5 interface with the file WAGE2.DTA open. The main window shows the results of the command `. corr wage educ exper tenure` (obs=526). The results are presented in a table format, highlighting the correlation coefficients between the variables wage, educ, exper, and tenure. A red box highlights the correlation matrix output.

Results
Mean estimation Number of obs = 526

	Mean	Std. err.	[95% conf. interval]
wage	5.896103	.1610262	5.579768 6.212437

end of do-file

```
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"
```

```
. corr wage educ exper tenure
(obs=526)
```

	wage	educ	exper	tenure
wage	1.0000			
educ	0.4059	1.0000		
exper	0.1129	-0.2995	1.0000	
tenure	0.3469	-0.0562	0.4993	1.0000

The right-hand pane shows the list of variables in the dataset, including smsa, northcen, south, west, construc, ndurman, trcommu, trade, services, profserv, profocc, clerocc, servocc, lwage, expersq, tenursq, educsq, and mean...educ. The bottom pane shows the Stata command log, which includes the command `. corr wage educ exper tenure` and the command `. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"`.

Correlations (cont'd)

- Using if: By adding a condition, such as **corr ... if profserv==1**, we can restrict the calculation to a specific subset, like seeing how education relates to wages specifically for those working in professional services.

StataNow/SE 19.5 — WAGE2.DTA

Results

```
. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"

. corr wage educ exper tenure if profserv==
invalid syntax
r(198);

end of do-file

r(198);

. do "/var/folders/r9/kzfnb9ps0fd_3y992brr918w0000gn/T//SD28840.000000"
```

```
. corr wage educ exper tenure if profserv==1
(obs=136)
```

	wage	educ	exper	tenure
wage	1.0000			
educ	0.4421	1.0000		
exper	0.0759	-0.2617	1.0000	
tenure	0.2266	-0.0588	0.5180	1.0000

```
.
end of do-file
```

Variables

Name	Label
smsa	=1 if live in SMSA
northcen	=1 if live in north central U.S
south	=1 if live in southern region
west	=1 if live in western region
construc	=1 if work in construc. indus.
ndurman	=1 if in nondur. manuf. indus.
trcommpu	=1 if in trans, commun, pub ut
trade	=1 if in wholesale or retail
services	=1 if in services indus.
profserv	=1 if in prof. serv. indus.
profocc	=1 if in profess. occupation
clerocc	=1 if in clerical occupation
servocc	=1 if in service occupation
lwage	log(wage)
expersq	exper^2
tenursq	tenure^2
educsq	
mean...educ	

Properties

Variables

Name

Lab1.do

```
88
89 * Create bar graph:
90 graph bar (mean) wage, over(educ)
91
92 * Create new variables:
93 gen educsq = educ^2
94 gen indicator = (educ>=12)
95 tab indicator
96
97 * Renaming variables:
98 rename indicator highschool
99
100 * Labeling variables:
101 label var highschool "1 if worker completed hi
102
103 * Bysort:
104 bysort highschool: tab wage
105 bysort highschool: sum wage
106
107 * Delete variable
108 drop highschool
109
110 * Compute mean:
111 mean wage
112 mean educ
113 mean exper
114
115 * Compute correlations:
116 corr wage educ
117 corr wage educ exper tenure
118 corr wage educ exper tenure if profserv==1
119
120 help summarize
121
122
123
```

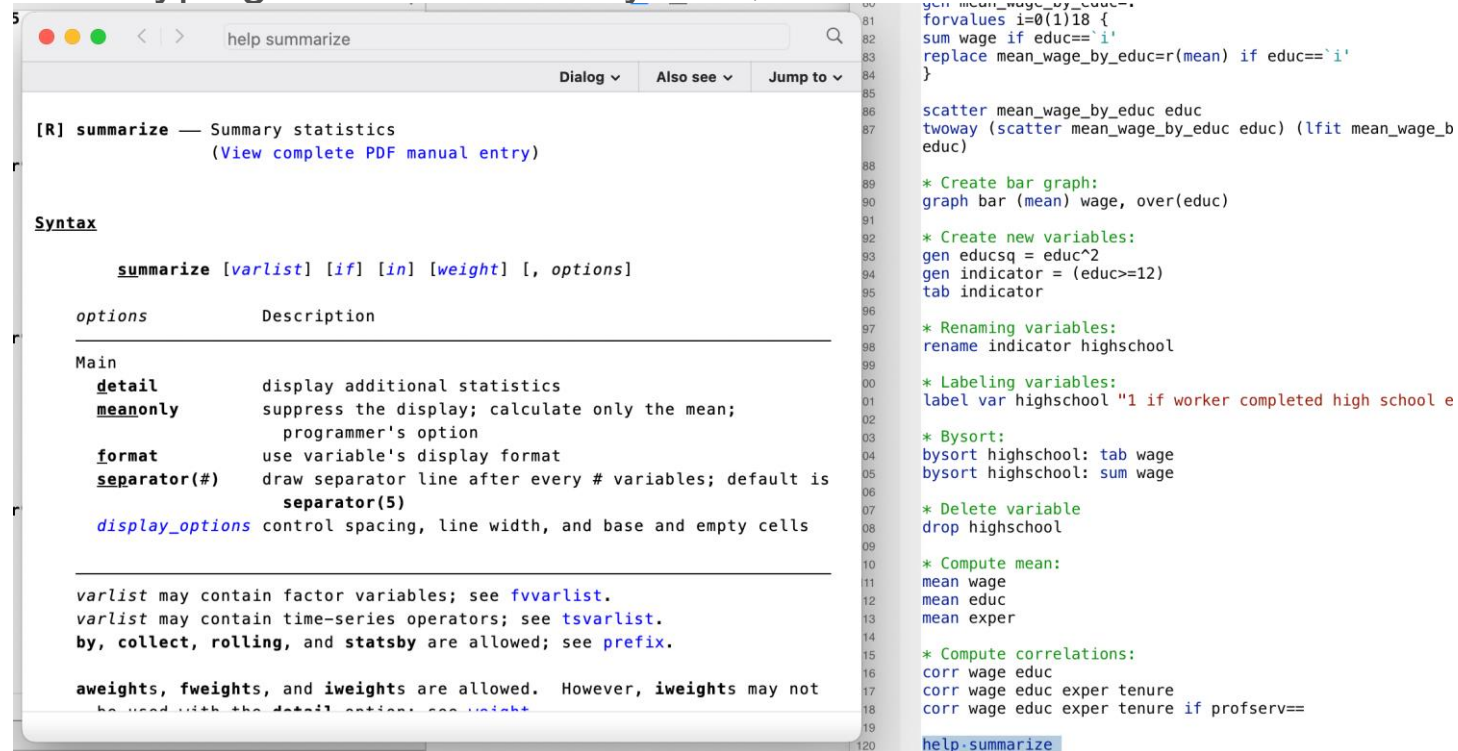
Append and Merge

- The commands **append** and **merge** can be used combine the current dataset with a second dataset. Use **append** to stack datasets vertically when you have the same variables but new observations (e.g., adding data from year 1998 to 1997). Use **merge** to link datasets horizontally when you have different variables for the same observations, matched by a key variable like *routeid* or *year*.
 - After a merge, Stata generates a system variable `_merge` to track which observations were matched; it is best practice to tabulate `_merge` immediately to verify the results.

```
47
48 // Suppose you have a separate file for another year
49 // --- APPEND: Adding observations (stacking) ---
50 use "data_1997.dta", clear
51 append using "data_1998.dta"
52
53 // --- MERGE: Adding variables (linking) ---
54 // Note: 'routeid' and 'year' must exist in both files
55 merge 1:1 routeid year using "extra_data.dta"
56 tabulate _merge
```

Help files

- Each Stata command has a help file that details the ways in which the command can be used. To pull up a help file, the syntax is '**help command name**' and after executing, a new window will appear. Note that in the syntax section, '**su**' is underlined. This indicates the minimum characters needed for Stata to recognize the command. So, instead of typing summarize every time, we can use **su** or **sum** or **summ**, etc.



The screenshot shows a Stata help window titled 'help summarize'. The window contains the following text:

```
[R] summarize — Summary statistics
      (View complete PDF manual entry)
```

Syntax

```
summarize [varlist] [if] [in] [weight] [, options]
```

options	Description
Main	
detail	display additional statistics
meanonly	suppress the display; calculate only the mean; programmer's option
format	use variable's display format
separator(#)	draw separator line after every # variables; default is separator(5)
display_options	control spacing, line width, and base and empty cells

varlist may contain factor variables; see [fvvarlist](#).
varlist may contain time-series operators; see [tsvarlist](#).
by, collect, rolling, and statsby are allowed; see [prefix](#).

aweights, fweights, and iweights are allowed. However, iweights may not be used with the detail option; see [weight](#).

On the right side of the window, there is a list of Stata commands that use the summarize command, including:

```
gen mean_wage_by_educ = ...
forvalues i=0(1)18 {
  sum wage if educ==`i'
  replace mean_wage_by_educ=r(mean) if educ==`i'
}

scatter mean_wage_by_educ educ
twoway (scatter mean_wage_by_educ educ) (lfit mean_wage_b
educ)

* Create bar graph:
graph bar (mean) wage, over(educ)

* Create new variables:
gen educsq = educ^2
gen indicator = (educ>=12)
tab indicator

* Renaming variables:
rename indicator highschool

* Labeling variables:
label var highschool "1 if worker completed high school e

* Bysort:
bysort highschool: tab wage
bysort highschool: sum wage

* Delete variable
drop highschool

* Compute mean:
mean wage
mean educ
mean exper

* Compute correlations:
corr wage educ
corr wage educ exper tenure
corr wage educ exper tenure if profserv==
```

Final Notes

- This slide deck is meant to serve as an introduction to Stata and I recommend perusing Stata's menus to see what else it's capable of. We will continue to learn other Stata commands during the semester, including how to run different kinds of regressions.
- When in doubt, read the help file! Whether it's an error message you keep getting or if you're unsure of how to use a command, the help file is the first place you should go.
- Please reach out to Professor Ren or TA (Ling) with any Stata questions you have!